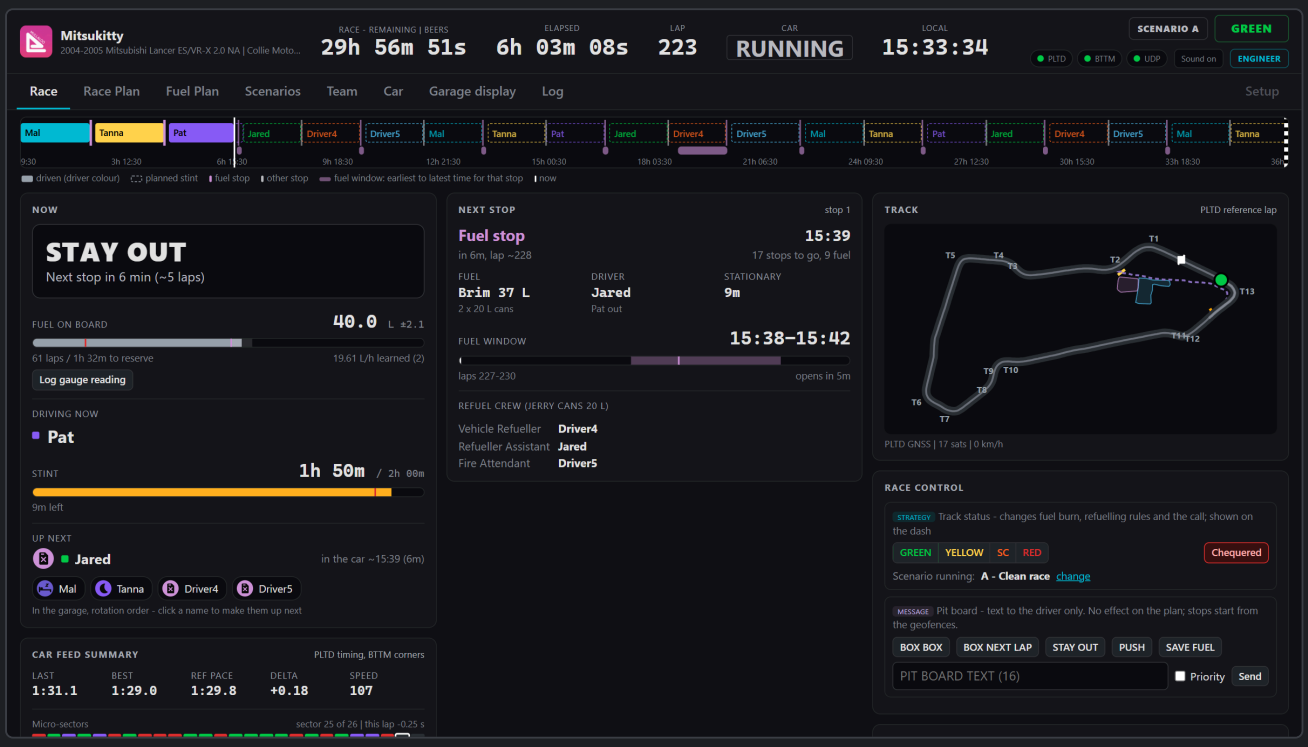
Pitwall Strategy Manager

User manual - endurance race strategy, live pit wall, fuel planning and crew management



Version 2.2.3 - test release

Mullaloo Motorsport Engineering - 2026 Pietro Gambarin

October 2026

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1. Introduction

Pitwall Strategy Manager is a self-hosted race engineering application for endurance racing. One laptop runs a small server; the engineer works in a browser on that laptop, and any phone, tablet or laptop on the same network can open the same screens view-only. It runs completely offline: no internet, no cloud account and no map tiles are needed at the track.

It continuously answers the questions a pit wall asks during a long race:

- **Should the car stop now?** A single call (STAY OUT, WINDOW OPEN, BOX IN 3 LAPS, NOTIFY DRIVER: BOX BOX...) recalculated every second.
- **How much fuel is in the car, and how far will it go?** A fuel model that learns from every brim fill and gauge reading.
- **What is the plan to the flag?** Stops, fuel loads and driver changes, kept legal against the selected regulations (stint limit, fuel stop minimum, mandatory stops, refuelling under flags).
- **What happens at each fuel stop?** Litres, jerry cans, crew by name and a step-by-step procedure for every fuel stop of the plan being run.
- **Who does what?** Driver rotation, crew presence, rest and sleep, refuel crew summoned by name before each fuel stop.
- **What happened?** Every change is written to an append-only log that can be replayed through the whole application.

Pitwall Strategy Manager is developed by Mullaloo Motorsport Engineering - Pietro Gambarin. It connects to the Mullaloo Motorsport Engineering devices in the car: the **PLTD** lap timer (laps, micro-sectors, pit lane geofences, pit board messages, flags, GPS track map) and the **BTTM** (brake and tyre temperatures, TPMS). Without them, everything can be entered by hand.

1.1 Who this manual is for

Reader	Read first
Testers evaluating the release	Sections 1.4 (what is new), 2, 3 and 4 (install, start, guided demo), then 12 (feedback)
Race engineer / strategist	Everything; sections 5 to 7 in detail
Refuel crew chief	Section 5.7 (Fuel Plan) and 6.5 (pit stop workflow)
Crew members using a tablet or TV	Section 3.3 (view-only access) and 5.11 (Garage display)

1.2 Versions 1.x and 2.x

Version 2 is a redesign of the user interface of the original version 1.x. Features, server, strategy engine and data files are the same; only the screens differ. Version 2 follows control-room design practice (ISA-101 "high performance HMI"): calm neutral screens, colour only when something needs attention, gauges with limits instead of bare numbers, and every piece of information in exactly one place. Both versions open the same race weekend files, but only one of them can run at a time.

Up to 2.1.4 the application was called **Pit Wall Pro2**. From 2.2.0 it is **Pitwall Strategy Manager**, and the "Pro1 / Pro2" names are retired. The version number lives in one place (`package.json`); the server console and the footer at the bottom of every page show it, for example *Pitwall Strategy Manager v2.2.3* | *Mullaloo Motorsport Engineering - 2026 Pietro Gambarin*.

1.3 Conventions

- **Bold names** are buttons, tabs or panel titles, for example **Race control**.
- `Monospace` is a file, command or value you type.

- Clock times are the laptop's local time, shown 24 hour (13:00, 00:00) by default or 12 hour with am/pm (1:00 pm), chosen in **Setup**. "Race time" means time since the green flag.
- Durations always carry their units so they cannot be misread: **1h 55m**, **7h 15m**, and under an hour **4m 52s**.
- NEW marks something added or changed since the 2.1.4 manual.

1.4 What is new since 2.1.4 NEW

Version	Change	Section
2.2.0	**Fuel Plan** tab: every fuel stop of the plan being run, with litres, jerry cans, refuel crew and a step-by-step procedure. Change the litres of a stop and preview where every later fuel stop moves.	5.7
2.2.0	**Stop handshake** replaces the four-step Stop gates: each step has a GEO signal (PLTD geofence) and an ENG signal (your button). New **FUEL ON** step.	5.5, 6.5
2.2.0	**+1 lap** button on the call: keep the driver out one more lap when the call says NOTIFY DRIVER: BOX BOX.	5.5, 7.5
2.2.0	**Refuel crew rules:** the outgoing driver goes straight to rest and is never on the crew; the incoming driver joins the fuel zone as Refueller Assistant, supervising hands-off.	7.6
2.2.0	**Shift draft** rebuilt stint by stint: next driver in, just out resting, one sleeping, at least two in the garage. Fuel crews come from the garage, so the plan no longer wakes anyone for a stop.	5.9, 7.6
2.2.0	Tabs: **Plan** is now **Race Plan**; **Fuel Plan** sits between it and Scenarios. Status bar: Elapsed is its own column, and Race, Elapsed, Lap, Car and Local all use the same large size.	5.3
2.2.0	Renamed Pitwall Strategy Manager; product, version and the Mullaloo credit in a footer on every page. New demo team (Mitsukitty).	1.2, 4.1
2.2.1	Race tab: the **Track map** moves to the top of the right column. **Race history** (was Race story) is shorter and scrolls inside its card.	5.5
2.2.2	Larger text (16 px base) on every tab. **Shifts & sleep** bars twice as tall, with high-contrast colours.	5.9
2.2.3	**Car** tab: Tyre sets at the top; live Track map and a new **Live timing (PLTD)** card with a micro-sector grid.	5.10

2. Requirements and installation

2.1 What you need

Item	Requirement
Host computer	Windows 10/11 laptop (Linux and macOS also work). Any recent laptop is enough; the reference machine is a Lenovo P52.
Software	Node.js 18 or newer (LTS). Install once from nodejs.org. Nothing else: all other components are inside the package.
Browser	Any current Chrome, Edge, Firefox or Safari, on the laptop and on viewing devices. The Race tab shows its three columns side by side on screens at least 1400 px wide; narrower screens stack them.
Network (optional)	For viewing devices and car telemetry: all devices on the same Wi-Fi network as the laptop (at the track: the VDG / MME_AP network, 192.168.0.x).

2.2 Install

1. Install **Node.js** (LTS) and accept the defaults.
2. Unzip the release (for example `mme-pitwall-strategy-manager-v2.2.3.zip`) to a folder you control, for example `C:\PitWall\pitwall-strategy-manager`. Do not run it from inside the zip.
3. That is all. The `node_modules` folder is included, so no download or internet connection is needed.

The folder contains:

File / folder	Purpose
`start-pitwall.bat`	Starts the server and opens the engineer screen.
`make-demo.bat`	Builds the demo race weekend (section 4).
`start-simulator.bat`	Starts the simulated PLTD and BTTM (section 4).
`data\`	Created on first run: race weekends, logs, car library, logo. Back this folder up.
`rules\`	Regulation profiles (B4B Events, BEERS).
`package.json`	Holds the version number shown in the console and the page footer.
`README.md`	Short technical notes and the change log.

> **Upgrading**
> Unzip the new version into a new folder and copy your existing `data` folder into it. Your weekends, cars and logo carry over. A 2.1.x `data` folder opens unchanged in 2.2.3.

3. Starting, stopping and access

3.1 Start

Double-click `start-pitwall.bat`. A console window titled *MME Pitwall Strategy Manager server* opens and prints the addresses, and the browser opens the engineer screen:

```
Pitwall Strategy Manager v2.2.3 - Mullaloo Motorsport Engineering - 2026 Pietro Gambarin
Engineer (this laptop):  http://localhost:8080
View-only devices:       http://192.168.0.23:8080
Data folder:             C:\PitWall\pitwall-strategy-manager\data
```

Leave the console window open for the whole event: closing it stops the server. If the server is restarted it resumes the last race weekend automatically, with nothing lost.

> **First start: Windows Firewall**
> Windows asks whether Node.js may use the network. Allow **Private networks**. If you refuse, the laptop still works but other devices and the car telemetry cannot connect.

3.2 Stop

Close the console window, or press **Ctrl+C** in it. Every change is already saved the moment it is made; there is no "save" step.

3.3 Engineer and view-only access

Device	Can do
The host laptop (any of its own addresses)	Everything. Badge in the top-right corner: **ENGINEER**.
Any other device	See every screen, run what-if scenarios locally, browse the log and replay. It cannot change anything (badge **VIEW ONLY**; the server rejects changes).

To let a second laptop make changes, add its IP address to `engineerHosts` in `data\settings.json` and restart. The server port (default 8080) is set in the same file.

4. Demo and simulator - guided tour

The fastest way to learn the application is the demo: a complete race weekend already six hours into a 36-hour race, with a simulated car feeding live data. Nothing in it touches your real weekends.

4.1 What the demo contains NEW

Item	Demo content
Event	**DEMO - BEERS 36h Collie**: BEERS standing regulations V4, 36-hour race on the Collie Motorplex long circuit (2.55 km, 13 turns, anti-clockwise)
Team and car	**Mitsukitty**, 2004-2005 Mitsubishi Lancer ES/VR-X 2.0 NA (45 L primary + 30 L secondary tank, jerry cans of 20 L)
People	6 drivers and 2 engineers: • **Mal** - owner, every role. • **Tanna, Pat, Jared** - drivers, mechanics, full fuel crew (refueller, assistant, fire), radio (Raceceiver relay), timing. • **Driver4, Driver5** - drivers, full fuel crew. • **Pietro, Rhys** - engineers: mechanic, timing, radio, race engineering. No driving, no fuel.
Shift plan	Drafted with the same **Draft shifts from the plan** button as on the Team tab (section 5.9): drivers rotate through drive, rest, sleep, garage and next in; the two engineers alternate in 6 h groups.
History	Practice session; race started 6 hours before the demo was built; two fuel stops done; Safety Car periods; car issues, a black-flag warning, an engineer off the wall for a 2 h break
Fuel model	Already calibrated from two brim fills (it learned about 19.6 L/h against a 18.75 L/h baseline)
Scenarios	Scenarios A-F created at the moment the demo was built; A running
Devices	Pointed at the simulator: PLTD at `127.0.0.1:8081`, BTTM at `127.0.0.1:8082`

4.2 Start the demo

1. Make sure the server is **not** running (close its console window).
2. Double-click `make-demo.bat`. It builds the demo weekend and makes it the active weekend. Run it again any time to get a fresh demo; it only replaces the DEMO weekend.
3. Double-click `start-pitwall.bat`. The engineer screen opens on the **Race** tab.
4. Double-click `start-simulator.bat`. A second console shows the simulated car: `PLTD sim: Pit Wall connected`, then a line per lap. Within a few seconds the device indicators in the status bar turn green and the car dot starts moving on the track map.

> **Why rebuild the demo?**
> The demo is anchored to the time it was built ("6 hours ago"), and its first planned fuel stop is only about 10 minutes away. If you come back later, run `make-demo.bat` again so the race clock and the plan make sense.

4.3 What the simulator does

The simulator pretends to be the two devices in the car, on this laptop, using the same network protocols as the real PLTD and BTTM firmware. The application cannot tell the difference.

Behaviour	Detail

Laps	About 92 s per lap with small random variation, reported exactly like the real PLTD (lap number, latched last lap time, live delta, speed, micro-sectors).
Track	The real Collie Motorplex long circuit (section 5.14), anti-clockwise through T1 to T13. In the pits the car follows the real pit lane: along the lane to the pits area, into the fuel area while fuelling, then out along the lane to the pit exit.
Laps through the pits	The pit lane passes the start/finish line, so the lap is counted where the car crosses it in the pit lane; the car rejoins at the pit exit without jumping.
Flags and driver pace	Yellow and Safety Car slow the car at once (not from the next lap), about 1.35x and 1.6x lap time; Red about 2.5x. SAVE FUEL on the pit board adds about 1.8 s per lap, PUSH takes about 0.6 s off, until the next stop.
Follows the engineer NEW	The simulated crew follows the stop handshake: it starts pouring when you press **FUEL ON**, fuels only if the plan or your entry says so, stops fuelling when you press **FUEL STOP COMPLETED** and sends the car out when you press **PIT OUT**. Untouched steps use default timings.
Obeys the pit board	Any pit board message containing BOX brings the car in at the end of the current lap (at least 20 s after the message). This is only to exercise the system: in the app, BOX BOX is a driver message with no effect on the plan. A message containing NO FUEL or DRIVER makes it a stop without fuelling.
Pit stop sequence	PIT ENTRY (25 s) - PITS (10 s, or 3 s after FUEL ON) - FUELLING (fuel minimum + 20 s) - PITS - PIT EXIT (15 s) - RUNNING. These are the same geofence states the real PLTD sends.
Driver acknowledgement	Every pit board message is "shown" on the dash and acknowledged by the driver 1.5 to 3.5 s later.
Brakes and tyres	Temperatures rise with speed and cool in the pits. Every 15 to 35 minutes the front-left brake spikes above 700 C for 8 seconds, to exercise the BTTM alarm and the automatic car issue.
TPMS	About 28.5 psi, rising slightly with tyre temperature.

4.4 Simulator options

Options can be added after the batch file name in a command window (for example `start-simulator.bat --fuelmin 1`), or by running `node tools\simulate.js` with the same options.

Option	Default	Effect
`--lap 92`	from the demo (92 s)	Base lap time in seconds.
`--fuelmin 8`	from the demo (8 min, BEERS)	How long the simulated crew spends fuelling, in minutes. Use 1 for quick tests; the application will then correctly flag the stop as under the regulation minimum.
`--startLap N`	last demo lap	Lap number to continue from.
`--pltd 8081` / `--bttm 8082`	8081 / 8082	Ports of the simulated devices. If you change them, change the device addresses in Setup to match.
`--pitwall URL`	`http://127.0.0.1:8080`	Where the server runs, if not on the default port.

4.5 Guided exercises

Work through these in order. Each takes a few minutes and covers one part of the application.

Exercise 1 - Read the Race screen (2 min)

1. Look at the status bar: time remaining, elapsed, lap, car state and local time, all in the same large type; scenario and flag on the right.
2. In the **Now** column read the call (normally STAY OUT with the time to the next stop), the fuel gauge and the driver's stint gauge.
3. In the **Next stop** column find what the next stop is, when, how much fuel, who gets in, and who is on the refuel crew.
4. Hover over **Fuel on board** for half a second: a tooltip explains it. Tooltips exist on most labels and settings.

Exercise 2 - Call the car in and run a pit stop (15 min)

1. Watch the call count down (BOX IN 3 LAPS, 2, 1) as the planned stop approaches; on the last lap it becomes **NOTIFY DRIVER: BOX BOX** with a **+1 lap** button beside it. Press **Send BOX BOX to the driver** (or **BOX BOX** in Race control at any time to stop sooner). The feed shows "Pit board: BOX BOX" with a MESSAGE tag, then the driver's acknowledgement. Nothing changes in the plan: it is only a message.
2. Within one lap the car crosses the pit-entry geofence: car state PIT ENTRY, the call becomes IN PITS, and the centre column becomes pit mode with the **Stop handshake** card on top. The Pit in row shows **GEO** lit and asks you to confirm: press **PIT IN**. The row reads HANDSHAKE.
3. The driver row is pre-selected with the planned driver (Pat out, Jared in). Press **DRIVER CHANGE DONE**.
4. When the crew starts pouring press **FUEL ON** (in the Fuel stop card). Fill in the **Fuel stop** card while they fuel: check the litres and cans, tick **Filled to brim**, tick the issues fixed. Watch the fuelling bar in the **Pit lane** card fill towards the 8-minute BEERS minimum. Open the **Garage display** and **Fuel Plan** tabs to see the crew views, then return to Race.
5. When fuelling ends press **FUEL STOP COMPLETED**, then **PIT OUT**. The card reads "All steps confirmed - waiting for the pit-exit geofence", and then "complete" once the car crosses the pit exit.
6. Repeat with another BOX BOX but press nothing: each step turns AUTO as the car moves past it, the planned fuel and driver are applied, and the stop is still complete.
7. Open **Race Plan**: both stops appear in the stint table with lane and fuelling times; only your confirmed brim fill was used as a fuel-model measurement.

Exercise 3 - Alerts and car issues (3 min)

1. In **Attention & feed**, press **Ack** on an alert. It leaves the attention list and moves to the race history.
2. Type an issue in "Log a car issue for the next stop" and press **Log**. It appears as a NEXT STOP job, and will be listed on the pit stop screen and on the Garage display.
3. Wait for (or look in **Car** for) a BTTM brake alarm: critical temperatures create a car issue automatically.

Exercise 4 - Safety Car (5 min)

1. Press **SC** in Race control. The flag in the status bar turns orange and the simulated car slows down.
2. Note the call: under BEERS rules refuelling is closed under Yellow/SC, so if a fuel stop is near the call says **STAY OUT - refuelling closed**. Under B4B rules (fuel area open under SC) the same situation gives **BOX NOW**.
3. Press **GREEN** to resume.

Exercise 5 - What-if (3 min)

1. Open **Race Plan**. In **What-if** enter a lap time change of +5 (rain) and press **Run**. The stint table and timeline show the scenario while it is active.
2. Try Green burn -8 %: see whether a fuel stop disappears. The **Drop one fuel stop** figure above the table gives the exact saving needed.
3. Press **Clear**. What-if never changes the real plan or the log.

Exercise 6 - Scenarios (5 min)

1. Open **Scenarios**. Compare the six bars: where the Safety Car and red flag sit, how many stops each needs, finish fuel, and the lap numbers above each bar.
2. On Scenario B press **Assumptions**. Drag the orange Safety Car to another time and drag its right edge to make it 30 minutes long. Watch the stop with the orange ring move into the Safety Car and the other stops re-plan.
3. Drag a stop on the bar: it and the stops before it get a cyan ring (pinned) and every later stop moves.
4. Drag the first fuel stop on Scenario A and drop it onto the next stop: the fuel moves there while both driver changes stay put.
5. Press **Run this scenario** on B. The status bar shows SCENARIO B; on Race the next stop and the timeline follow it, with its Safety Car drawn on the timeline.
6. Press **Reset** on B, then **Run this scenario** on A to go back.

Exercise 7 - Fuel Plan (5 min) NEW

1. Open **Fuel Plan**. The timeline shows only the fuel stops of the plan being run: F1 and F2 done, F3 next, the rest planned.
2. Click F5. The right-hand card shows litres on arrival, litres to add and on leaving, the jerry cans to bring, the refuel crew and who else attends. The step-by-step card on the left gives the procedure with times and names.
3. Under **Change the litres for F5** press **-5** twice. The preview shows every later fuel stop moving earlier, the new fuel stop count and finish fuel. Press **Cancel**, or **Apply to the running plan** to re-plan Scenario A.
4. If you applied it: open **Scenarios** and press **Reset** on A to undo.

Exercise 8 - A second device (3 min)

1. On a phone or tablet on the same Wi-Fi, open the "View-only devices" address from the console.
2. The badge shows VIEW ONLY; controls are hidden. Choose **Garage display** for the big-type crew screen.
3. Make a change on the laptop (for example a flag) and watch it appear on the second device within a second.

Exercise 9 - Replay (3 min)

1. Open **Log** and press **Start replay**. A purple banner shows the replay time.
2. Drag the slider, or press **Play** at 60x. Switch to **Race** and **Race Plan**: they show exactly what the pit wall knew at that moment.
3. Press **Back to live**.

4.6 Leaving the demo

Your own weekends are never changed by the demo. To switch back: **Log** tab, **Weekend files**, press **Open** next to your weekend. To start a real weekend, open **Setup** and press **Start a new weekend instead** (section 6). Close the simulator window when you are done with it.

5. Screen reference

5.1 How to read the screens

The application uses colour sparingly so that anything coloured means something:

Colour	Meaning
Grey / white	Normal. No action needed.
Amber	Warning: something is approaching a limit (fuel window open, stint ending soon, brake in the red band).
Red (pulsing if unacknowledged)	Critical: act now (box this lap, fuel at reserve, over a limit).
Lilac	Fuel: fuel stops, fuel windows, planned fuel arrival.
Cyan	Strategy items, pinned stops, the selected stop, primary buttons.
Flag colours	The track flag in the status bar (green outline, yellow, orange Safety Car, red).
Driver colours	Each driver keeps one colour on the timeline, the stint table and the shift plan.

Gauges are bars against their limit: the red tick is the limit (reserve fuel, 15-minute stint warning), the lilac tick a planned value, the white line "now".

5.2 What changes the strategy, and what does not

Everything you can do falls in one of three groups. In the feed and the log, items that change the strategy carry a cyan **STRATEGY** tag, and driver messages a lilac **MESSAGE** tag.

Group	Examples	Effect on the plan
STRATEGY - changes the timeline	Geofence events and the stop handshake (pit in, driver change, fuel on, fuel stop, pit out); track flag (Green, Yellow, SC, Red); **+1 lap**; driver corrections and rotation; gauge readings; litres set on the Fuel Plan; switching or editing a scenario; session start/end	The plan, the call, fuel estimate, driver times and crew summons are recalculated immediately.
MESSAGE - to the driver only	Pit board: BOX BOX, BOX NEXT LAP, STAY OUT, PUSH, SAVE FUEL, custom text	None. The text is shown on the PLTD dash and logged with the driver's acknowledgement. Pressing BOX BOX does not create a stop: the stop starts when the car actually crosses the pit-entry geofence.
Information	Car issues, penalties, crew status, tyre measurements, notes	No direct effect on stop timing (crew status and driver availability do change who is assigned and the rotation).

> **Why flags are strategy, not messages**
> A track flag changes the strategy because the rules and the car change: under Safety Car or Yellow the car burns less fuel, under BEERS rules refuelling is closed, and the call may become BOX NOW. The flag is also shown on the PLTD dash, but that is a side effect.

5.3 Status bar

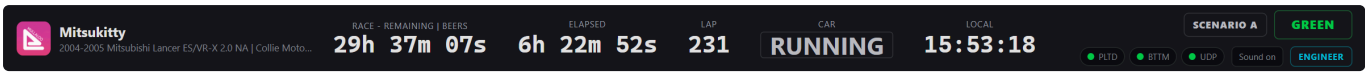


Figure 1 - Status bar

Element	Meaning
Team and car	Team name, car number, car, track. Logo top-left.
Race - remaining	Race time remaining; before the start, time to the planned start. The label also names the session and the regulations.
Elapsed NEW	Time since the green flag, in its own column. Race, Elapsed, Lap, Car and Local all use the same large type and line up.
Lap / Car	Current lap and car state from the PLTD geofences: RUNNING, PIT ENTRY, PITS, FUELLING, PIT EXIT.
Local	The laptop's clock.
SCENARIO	Which scenario is running (A unless you switch in Scenarios). Amber when it is not A. Click it to open the scenarios.
Flag	Current track flag (set in Race control; mirrored to the PLTD dash).
PLTD / BTTM / UDP	Device connection: green connected, amber connecting or stale data, red offline, dark disabled.
Sound on/off	Mutes the alert chime on this device.
ENGINEER / VIEW ONLY	What this device is allowed to do (section 3.3).

A red or amber line under the status bar means the current call is critical or a warning. At the bottom of every page a footer shows the Mullaloo logo, the product name and version, and the credit line.

5.4 Timeline

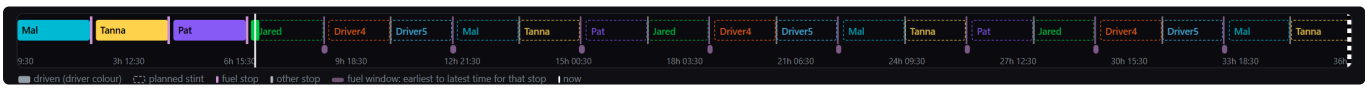


Figure 2 - Race timeline with legend

The whole race from green flag to chequered flag. Solid blocks are stints already driven (in driver colour, striped = current); dashed blocks are planned stints. Lilac ticks are fuel stops, grey ticks other stops. The lilac bar under a planned fuel stop is its **fuel window**: the earliest to the latest time that stop can be made (hover it for the exact times and laps). A wide window means the stop can move a lot, for example to take advantage of a Safety Car; a narrow one means it cannot. The white line is now.

5.5 Race tab

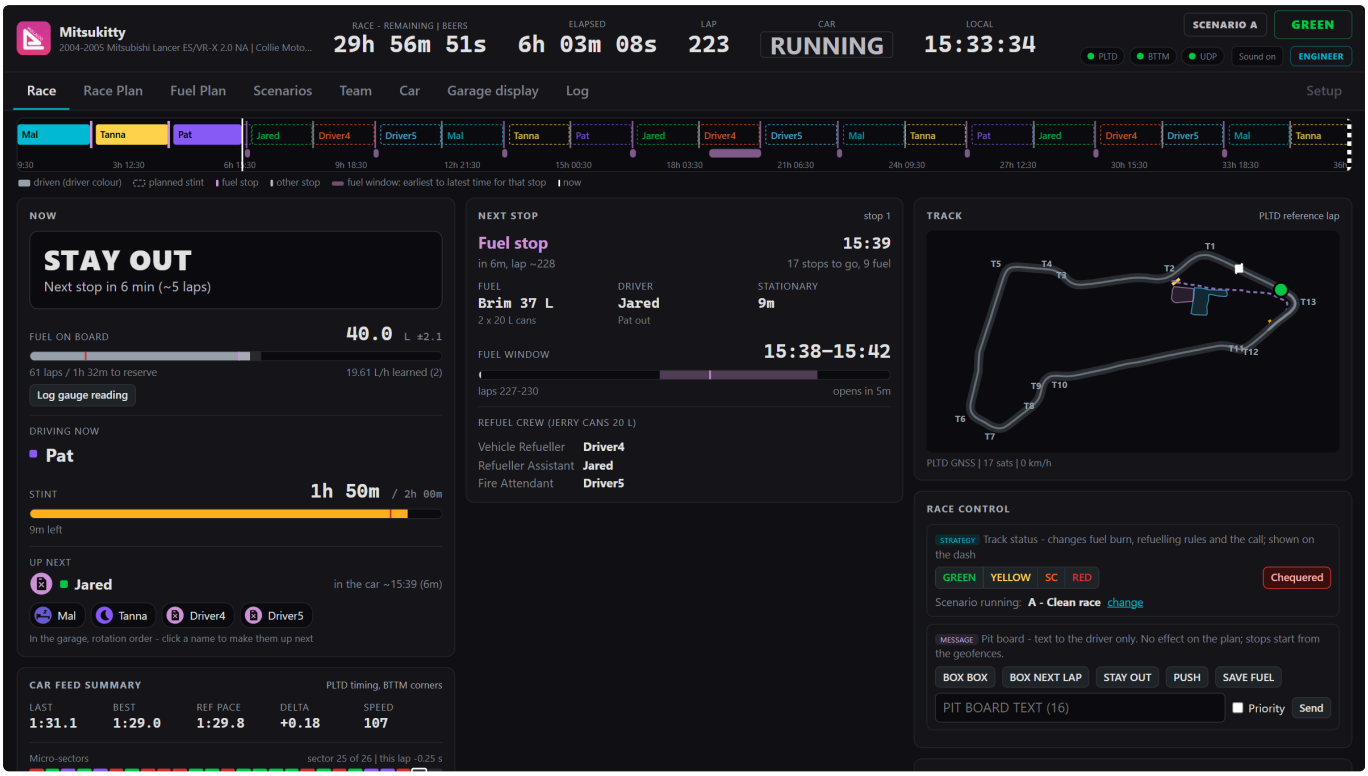
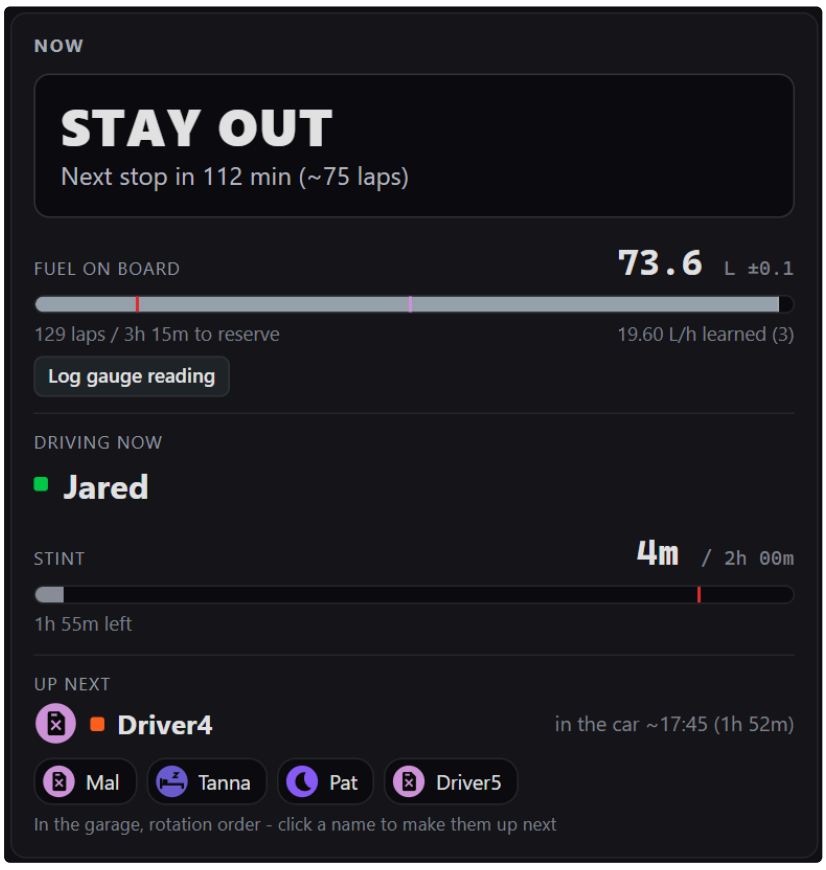


Figure 3 - Race tab: Now, Next stop, and the Track map, Race control and feed on the right

The overview. Three columns, each with one job: **Now** (left, with the Car feed summary under it), **Next** (centre), and on the right the **Track** map NEW, **Race control** and **Attention & feed**.

Now

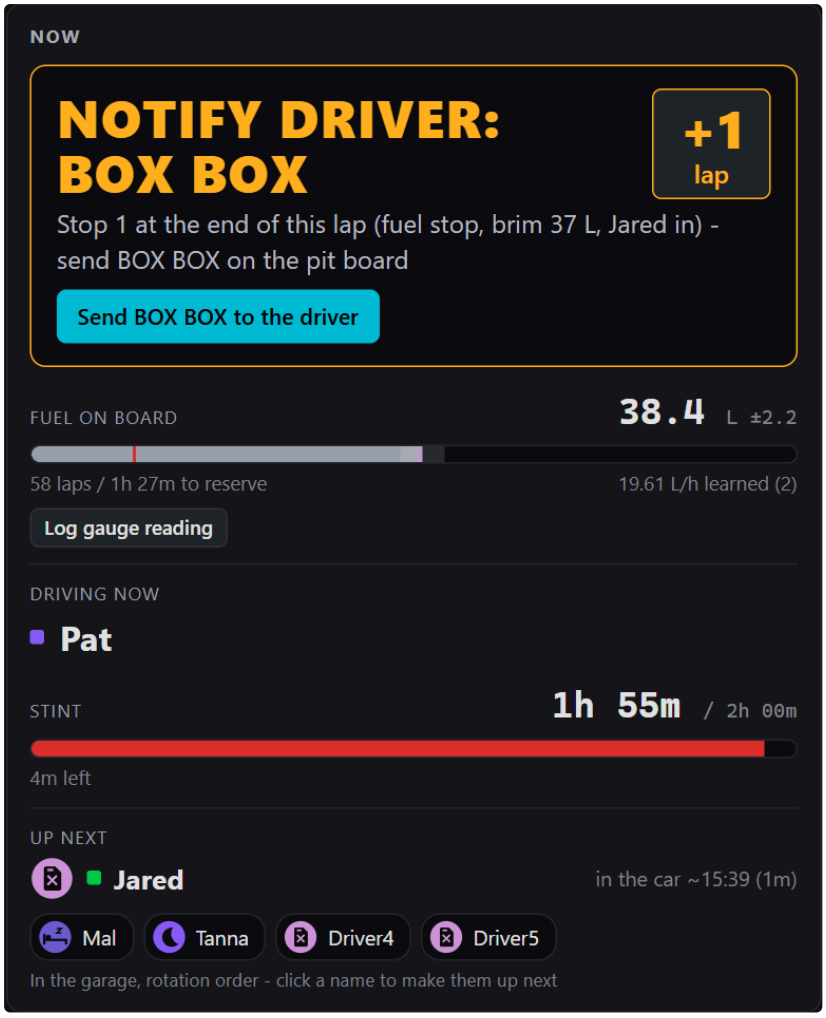


Now column

- **The call** - the one instruction to give. See Appendix B for every call and its meaning.
- **Fuel on board** - estimated litres with uncertainty (the light band). Red tick: reserve floor. Lilac tick: planned arrival at the next fuel stop. Underneath: laps and time to reserve, and the burn rate in use (baseline or learned).
- **Log gauge reading** - enter what the fuel gauge or dipstick shows; this re-anchors the estimate and calibrates the model.
- **Driving now** and **Stint** - continuous time behind the wheel against the regulation maximum; amber at 15 minutes left, red at 5.
- **Up next** - the next driver and the planned time they get in. Below, the rest of the rotation with presence icons; the engineer can click a name to make that driver up next.

Icon	Meaning
Steering wheel (cyan) NEW	Driving
Jerry can (lilac)	Available, and on the refuel crew for the next fuel stop
Spanner (grey)	Available in the garage (mechanic), ready for work on the car
Tick (green)	Available
Bed (purple)	Sleeping
Moon (violet)	Resting
Flag (orange)	On flag duty
Cross / red bar	Offline / unavailable

The call near a stop and +1 lap NEW

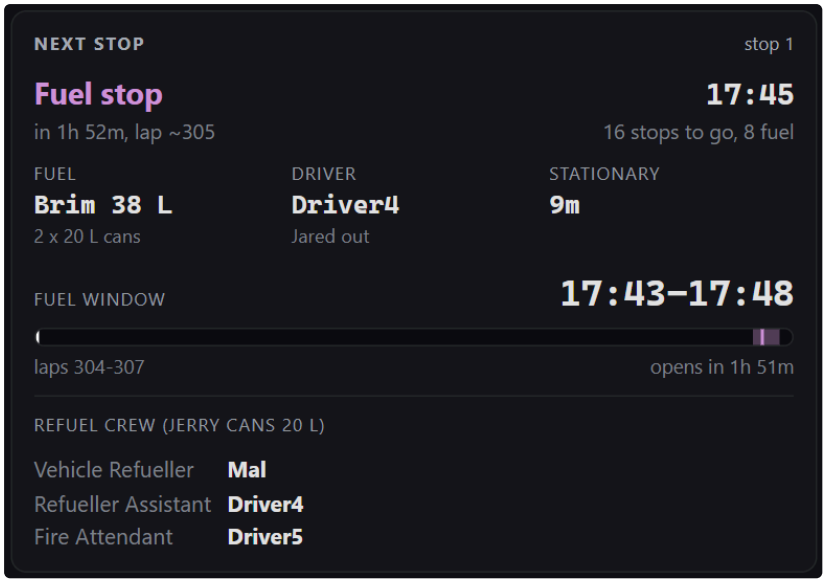


Last lap of the stint: NOTIFY DRIVER: BOX BOX, with Send BOX BOX and the +1 lap button

From 5 minutes before the planned stop (the end of the driver's allocated stint) the call counts laps: BOX IN 3 LAPS, 2, 1. On the last lap it becomes **NOTIFY DRIVER: BOX BOX** with a notification and a one-click **Send BOX BOX to the driver**. Once sent, it reads **BOX THIS LAP** with the time sent and whether the driver has acknowledged.

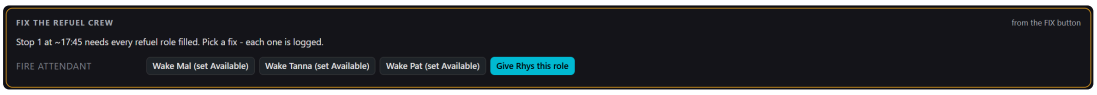
+1 lap keeps the driver out one more lap. The called stop moves one lap later in the running plan and in every scenario; its fuel goes up by one lap's burn, the planned driver change is kept, and the call and the log show "stint extended +1 lap". Press it again for a further lap. The extension clears when the car next enters the pits. The button is disabled, with the reason in its tooltip, if the extra lap would bring the fuel below the reserve or the stint over the maximum.

Next stop



Next stop

Everything about the next planned stop: type and clock time, time and laps to it, stops still to go, fuel to add (and cans), driver change, planned stationary time, the fuel window as a bar against now, and the refuel crew by name. Crew are assigned only from members tagged with that role and currently Available, and never the outgoing driver; the incoming driver is listed as Refueller Assistant (supervising) - see section 7.6. If nobody qualifies, the slot shows **UNASSIGNED** with the reason, who could help, and a **FIX** button. **FIX** opens the Team tab at **Fix the refuel crew**, which offers one-click fixes: wake a tagged person who is resting or sleeping (set them Available), or give an available person the missing role. Each fix is logged.



Fix the refuel crew (Team tab), opened from FIX

When the PLTD is offline, manual car state buttons appear here.

Pit mode and the stop handshake NEW

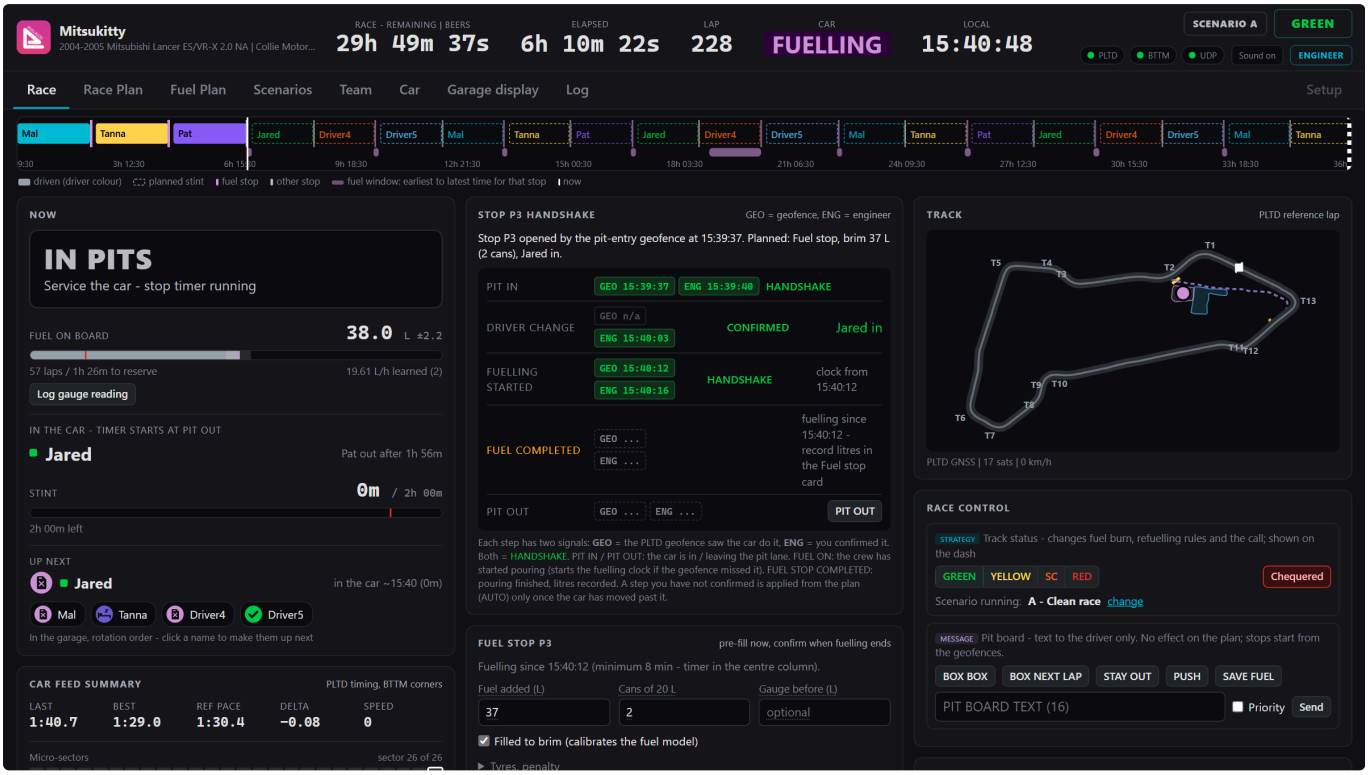


Figure 4 - Pit mode: the centre column becomes the pit stop screen while the car is in the pit lane

When a stop is due, or the PLTD pit-entry geofence opens one, the centre column becomes the pit stop screen, top to bottom:

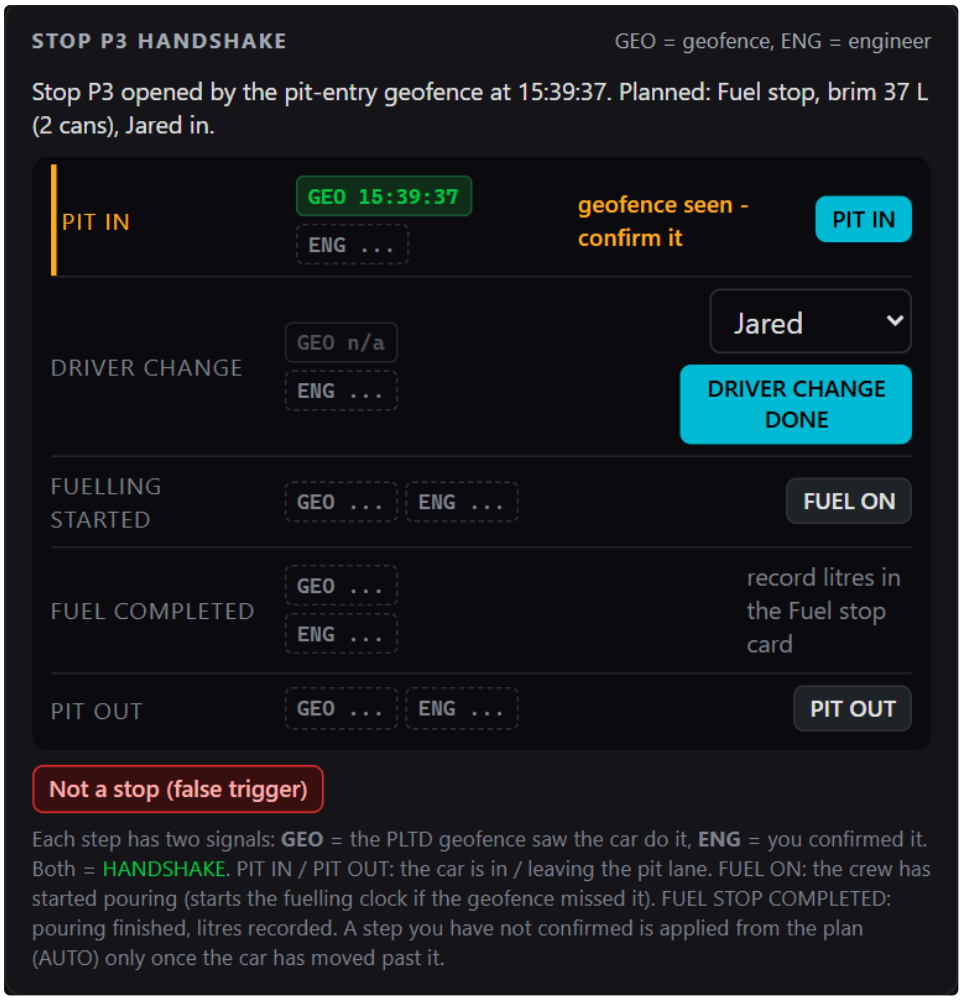
- **Stop handshake** - the five steps of the stop (below).
- **Fuel stop** - the fuel entry, available from pit in: litres and cans pre-filled from the plan, gauge before, **Filled to brim**, tyres, penalty served, issues fixed; buttons **FUEL ON**, **FUEL STOP COMPLETED** and **No fuel this stop**.
- **Pit lane** - fuelling time from the PLTD fuel-bay geofence against the regulation minimum (B4B 5 min, BEERS 8 min), time since pit entry against the mandatory-stop minimum where the rules have one (B4B: 5 min), what was planned at pit in and what was recorded, the refuel crew and the jobs to check at this stop. **Override car state (geofence missed)** is folded away under the card.



Pit lane card: fuelling against the regulation minimum, planned and recorded, jobs and refuel crew



Stop due, car not in yet: the PLTD opens the stop by itself; PIT IN opens it by hand



Stop handshake as the car enters: the pit-entry geofence (GEO) has seen it, the engineer confirms with PIT IN

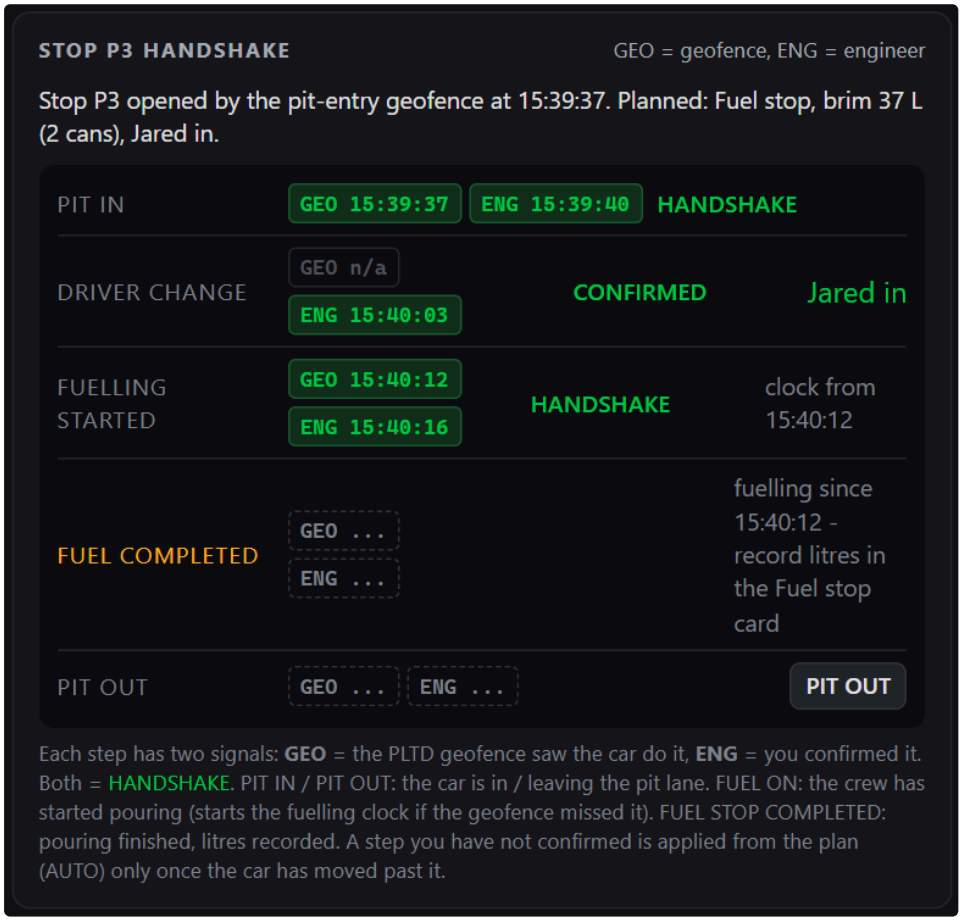
Every step has two signals. **GEO** means the PLTD geofence saw the car do it; **ENG** means you confirmed it. Both together read **HANDSHAKE**. There are no countdowns: you press each step when you see it done.

Step	GEO	ENG (what you press)
Pit in	pit-entry geofence	**PIT IN**: the car is in the pit lane. If the PLTD missed the pit entry, this also opens the stop by hand.
Driver change	none	Choose who gets in (pre-selected from the plan) and press **DRIVER CHANGE DONE**.
Fuelling started	car in the fuel zone	**FUEL ON**: the crew is pouring. Starts the fuelling clock if the geofence missed it; works with the PLTD connected too.
Fuel completed	car left the fuel zone	**FUEL STOP COMPLETED** in the Fuel stop card: litres and cans recorded. **No fuel this stop** for a driver-only stop.
Pit out	pit-exit geofence	**PIT OUT**: the car is leaving the pit lane.

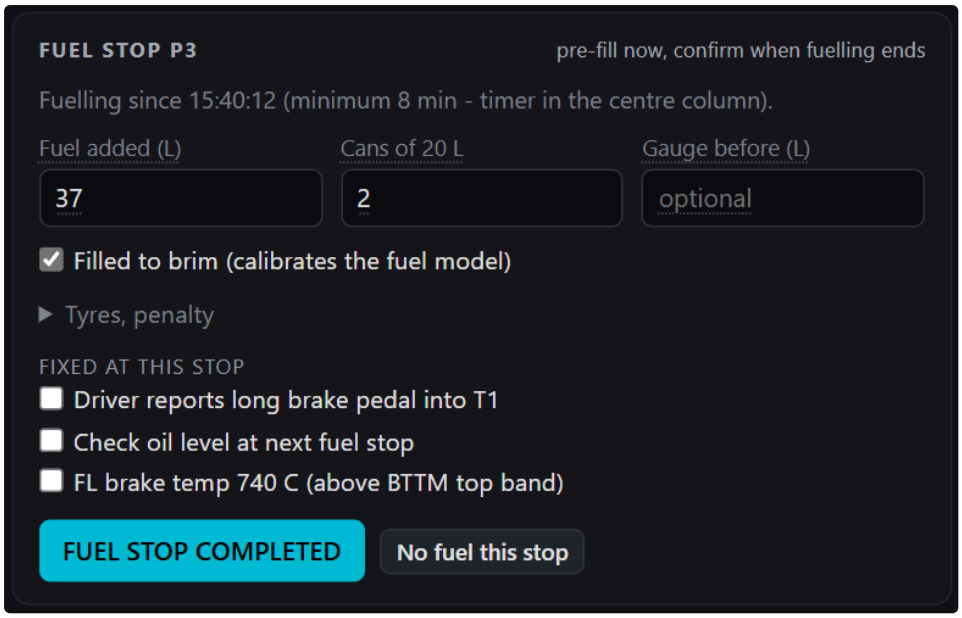
Reading the handshake:

- Amber "geofence seen - confirm it": the car did it and you have not confirmed yet.
- "Waiting for the geofence": you confirmed and the car has not got there yet.
- A step nobody confirms is still applied from the plan (**AUTO**) once the car has moved past it: the planned driver is set and the planned litres are recorded as added (not used to calibrate the fuel model). All steps are logged with STRATEGY tags.
- **Not a stop (false trigger)**, under the steps, rejects a false geofence event; the stop is removed from the strategy.

- If the fuel stop was applied automatically, **Correct it** reopens the card; **SAVE CORRECTION** replaces the automatic entry.
- With the PLTD offline the buttons also set the car state, so the pit lane and fuelling timers keep working. With the PLTD online the geofences drive the car state and the buttons confirm the steps. If a geofence is missed entirely, **Override car state** under the Pit lane card sets it by hand.
- The regulation timers (fuelling minimum, mandatory stop minimum) are in the Pit lane card; they are rules, not step countdowns. Once **No fuel this stop** is recorded, the fuelling minimum no longer applies and the fuel-bay geofence is ignored for that stop.
- After **PIT OUT** the card says "All steps confirmed - waiting for the pit-exit geofence"; the stop is **complete** only once the car crosses the pit exit. If no fuel was recorded but the PLTD still shows the car in the fuel bay, a note says so; if the car has really left and the geofence missed it, use **Override car state**.



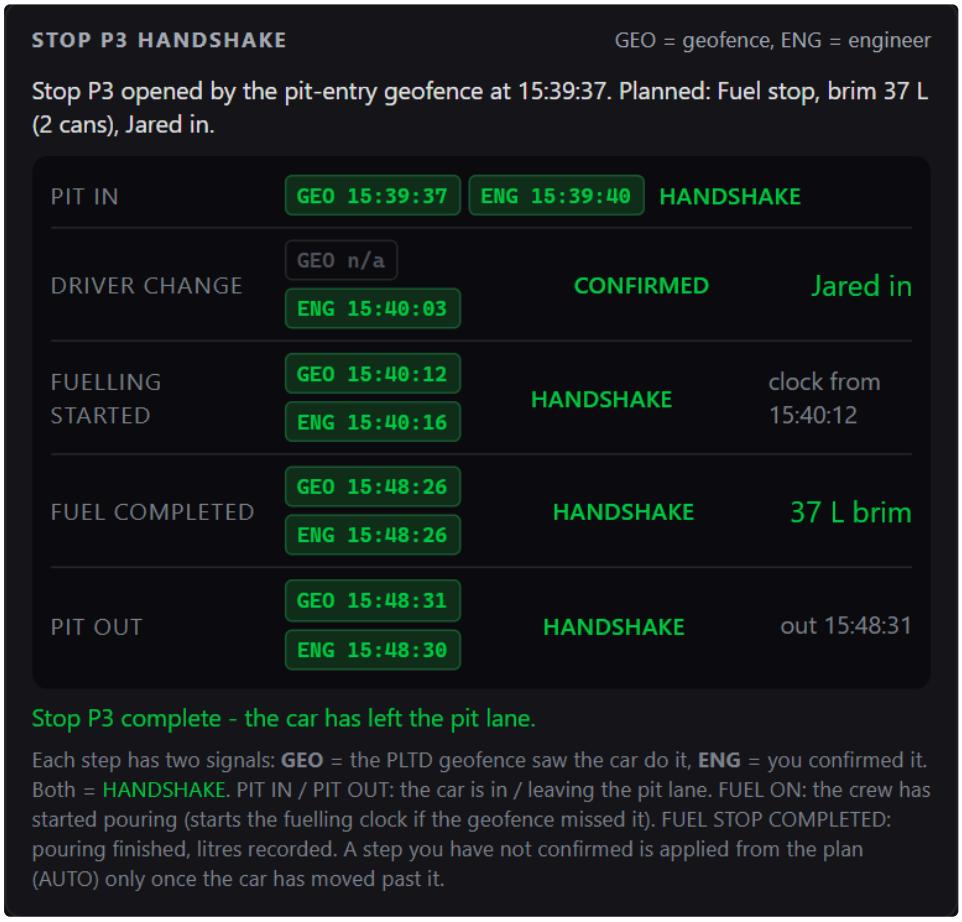
During fuelling: Pit in and Fuelling started are HANDSHAKES, the driver change is CONFIRMED (it has no geofence), fuel completed and pit out still to come



Fuel stop card, available from pit in: pre-fill it while fuelling, then press FUEL STOP COMPLETED

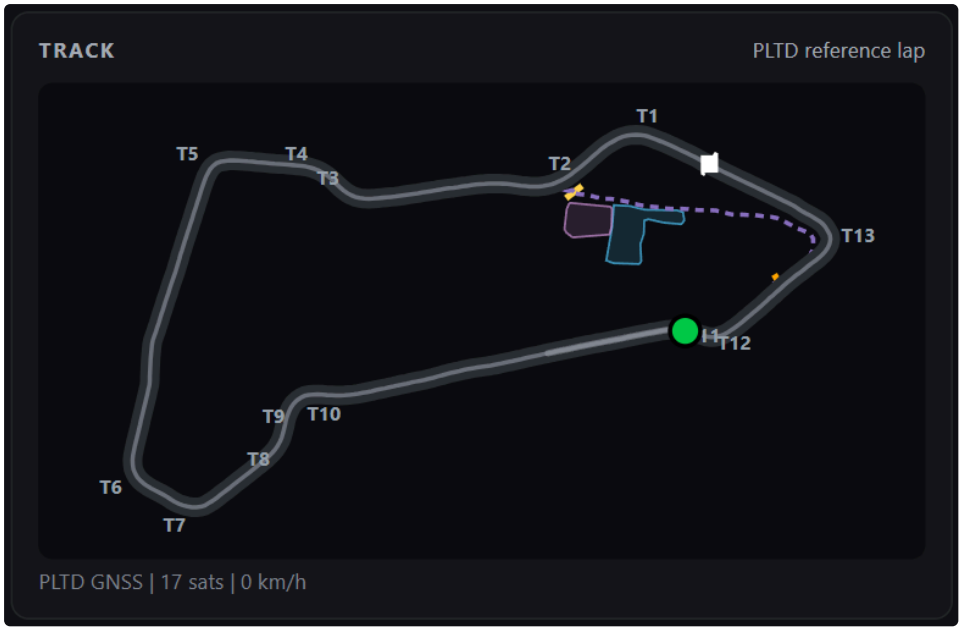
Brim fills make the fuel model accurate

Whenever the tank is filled until full, tick **Filled to brim**. Each brim fill is an exact measurement of how much fuel the last stint used, and the model recalibrates from it.



A stop driven by the engineer: every step a HANDSHAKE, stop complete after the pit-exit geofence

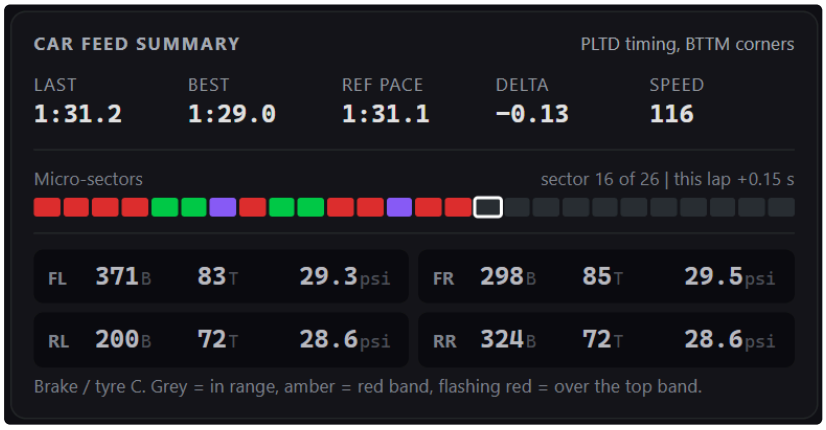
Track map and Car feed summary



Track map (top of the right column)

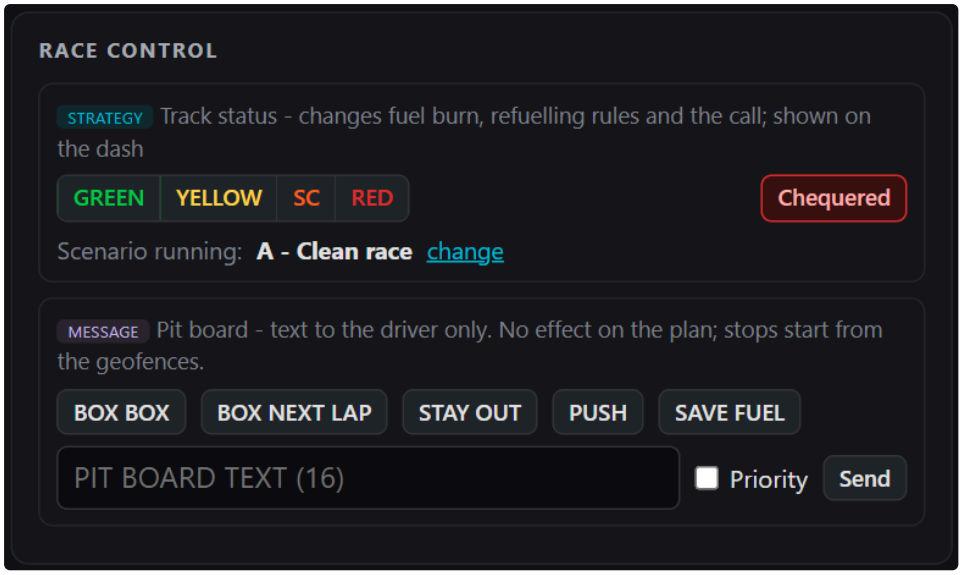
The track map sits at the top of the right column NEW. It shows the circuit with its turn numbers (for Collie, T1 to T13 of the long circuit, section 5.14), the start/finish line and the pit zones; the dot is the car (coloured by car state, grey if the position is old) with a short trail. Until the PLTD has recorded its own reference lap the outline comes from the track preset; after that the PLTD's recorded lap is used, which is exact for the car's own GPS.

Car feed summary (left column, under Now) gives last lap, best lap, reference pace (median of recent clean green laps), live delta and speed; the micro-sectors of the current lap from the PLTD (as many as its recon lap detected: green faster, red slower, purple best, grey not yet run, white outline = the sector the car is in, with the lap delta so far); and the four corners from the BTTM: brake and tyre temperature and TPMS pressure. Corner values are grey when in range, amber in the BTTM red band and flashing red above the top band. The Car tab has the same timing in more detail (section 5.10).



Car feed summary with micro-sectors

Race control (engineer only)

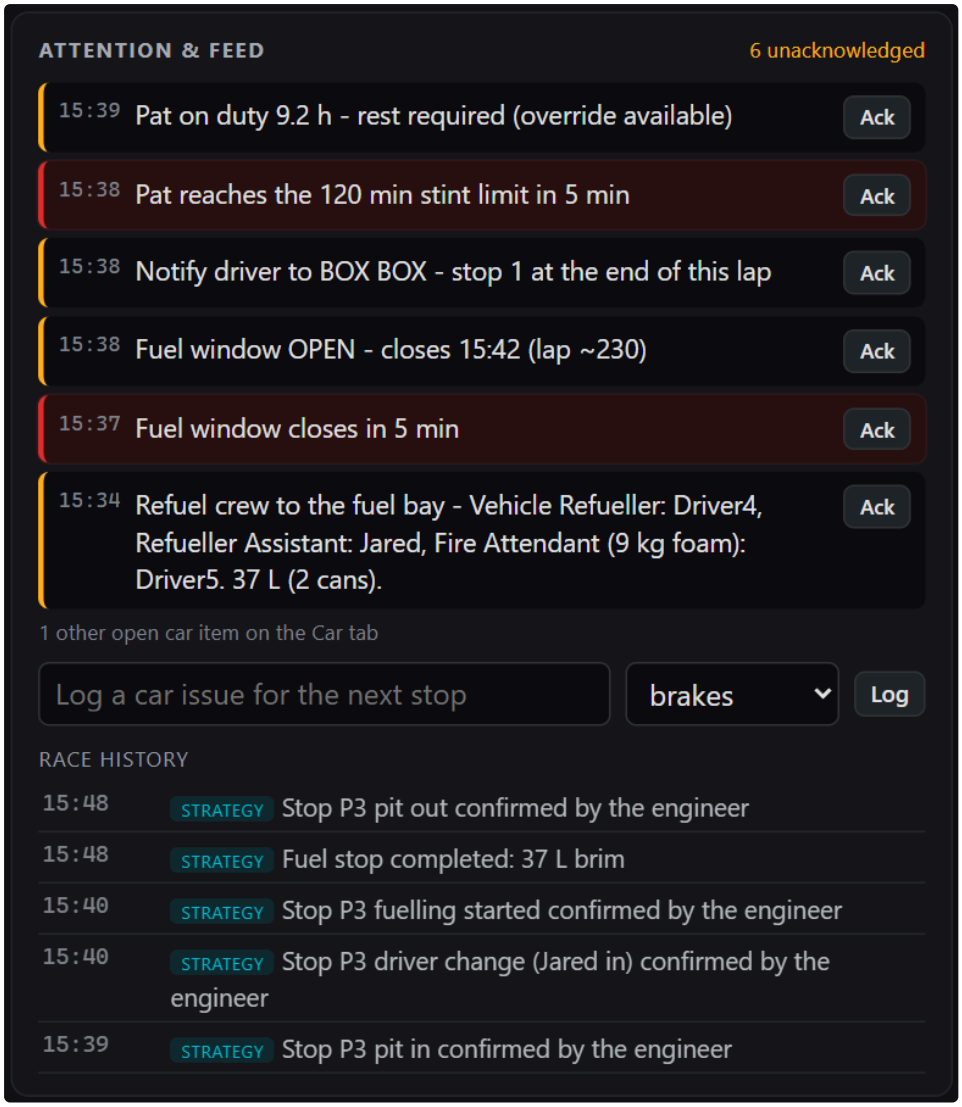


Race control

Before a session: choose the session, start fuel, starting driver and tyre set, and press **Green flag - start session**. During a session the panel has two clearly separated groups:

- **STRATEGY - Track status:** the track flag (changes fuel burn, refuelling rules and the call; also shown on the dash). Changing straight from one flag to another (for example Yellow to SC) is logged as one change. **Chequered**, and which scenario is running with a link to change it.
- **MESSAGE - Pit board:** presets or up to 16 characters of your own, shown on the PLTD dash. **Priority** keeps the message up until the driver acknowledges it. These never change the plan.

Attention & feed



Attention & feed, with the Race history below

The top of the feed is what needs attention: unacknowledged warnings and critical alerts (press **Ack**) and jobs for the next stop (press **Fixed** when done). Below is the **Race history** NEW (was Race story): flags, stops, pit board messages and acknowledgements, issues, penalties, crew status changes and acknowledged alerts, newest first. The history is a short window (about a quarter of the screen height) that scrolls inside its card, never the page, and shrinks further only to stay level with the taller of the other two columns. Alerts also pop up briefly in the bottom-right corner with a chime.

5.6 Race Plan tab

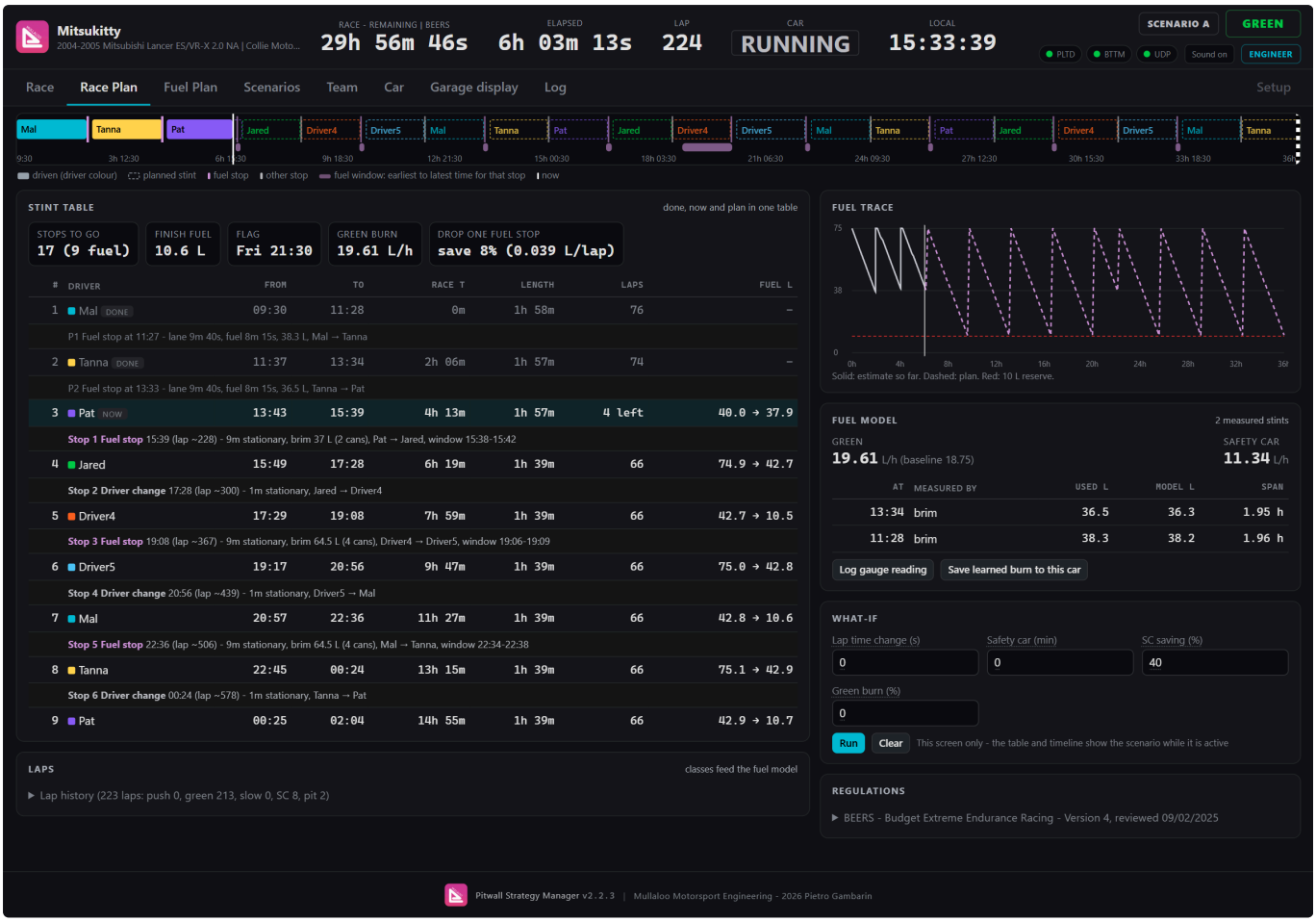


Figure 5 - Race Plan tab (was Plan)

- **Summary** - stops to go (and how many are fuel stops), projected fuel at the flag, flag time, green burn rate, mandatory stops counted, and **Drop one fuel stop**: the fuel saving the planner actually needs to remove one fuel stop.
- **Stint table** - one table for the whole race: stints already driven (grey, with each completed stop and its lane time, fuelling time, litres and compliance), the current stint (highlighted), and the plan to the flag with every stop inline (time, lap, stationary time, fuel, driver change, window).
- **Laps** - lap history with each lap's class: push, green, slow, SC, pit.
- **Fuel trace** - fuel level over the race: solid so far, dashed planned, red reserve line.
- **Fuel model** - current green and Safety Car burn rates, every measurement used to learn them, **Log gauge reading**, and **Save learned burn to this car** (stores the learned rates as the baseline for this car at this track for future events).
- **What-if** - lap time change, a Safety Car of N minutes with a fuel saving, or a green burn change. Local to the screen; the stint table and timeline show the scenario while it is active.
- **Regulations** - the profile in force and its key rules.

5.7 Fuel Plan tab NEW

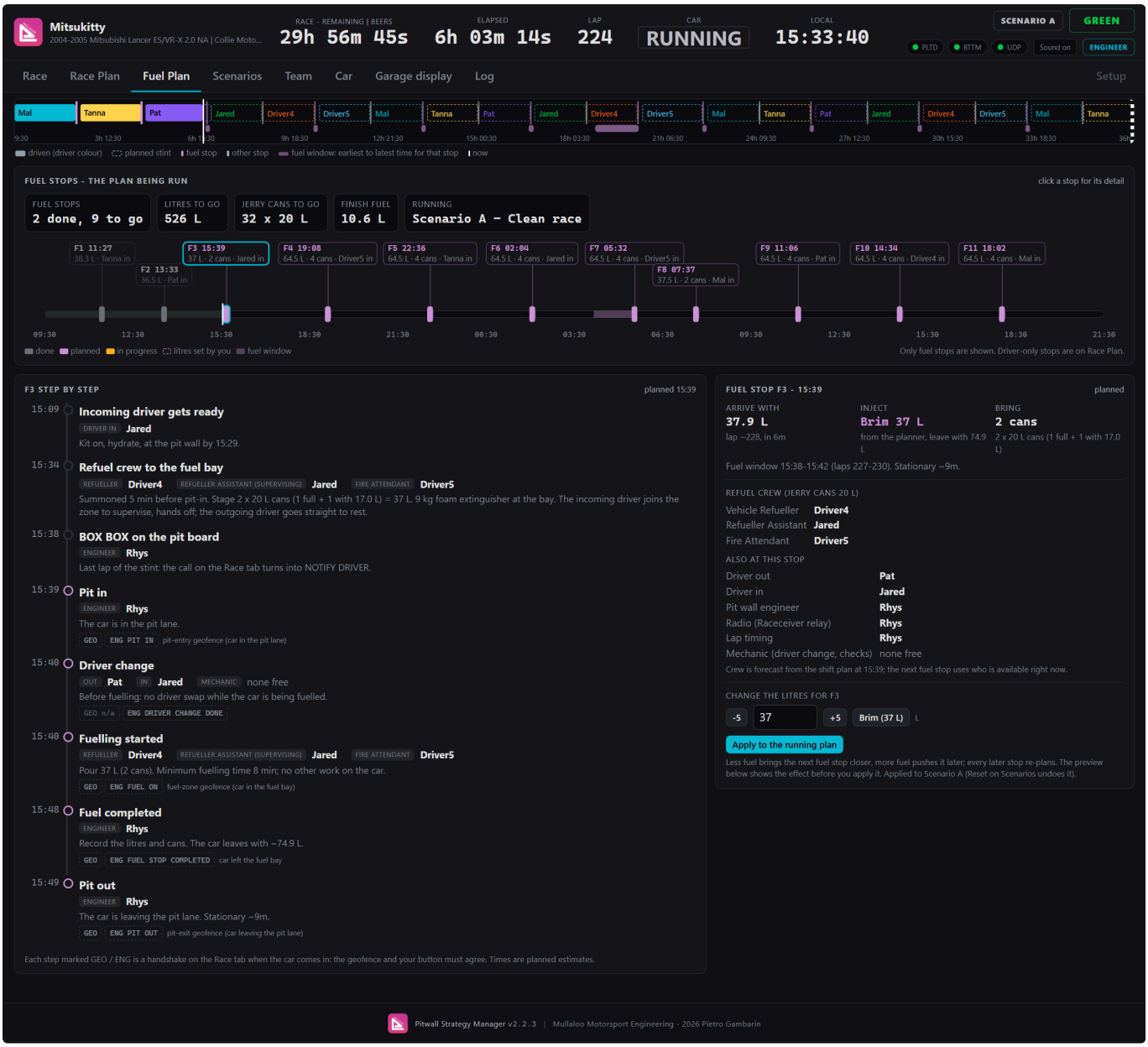
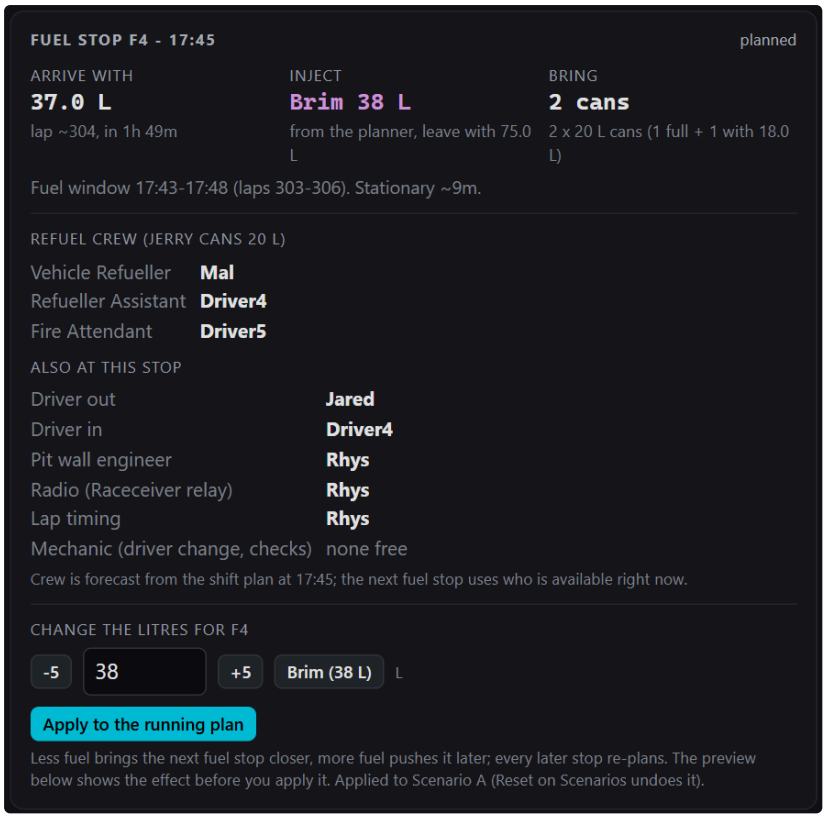


Figure 6 - Fuel Plan tab: the fuel stops of the plan being run, the selected stop's procedure and its detail

The Fuel Plan is the refuel crew's view of the race. It shows only the **fuel stops** of the plan being run (the running scenario); driver-only stops stay on Race Plan.

Timeline. Above it: fuel stops done and to go, litres to go, jerry cans to go, finish fuel and the scenario running. Each stop on the line has a short label - number, time, litres, cans and the driver coming in - and a marker: grey done, lilac planned, amber in progress, dashed outline when you set its litres, with its fuel window underneath. Click a stop (label or marker) to select it.

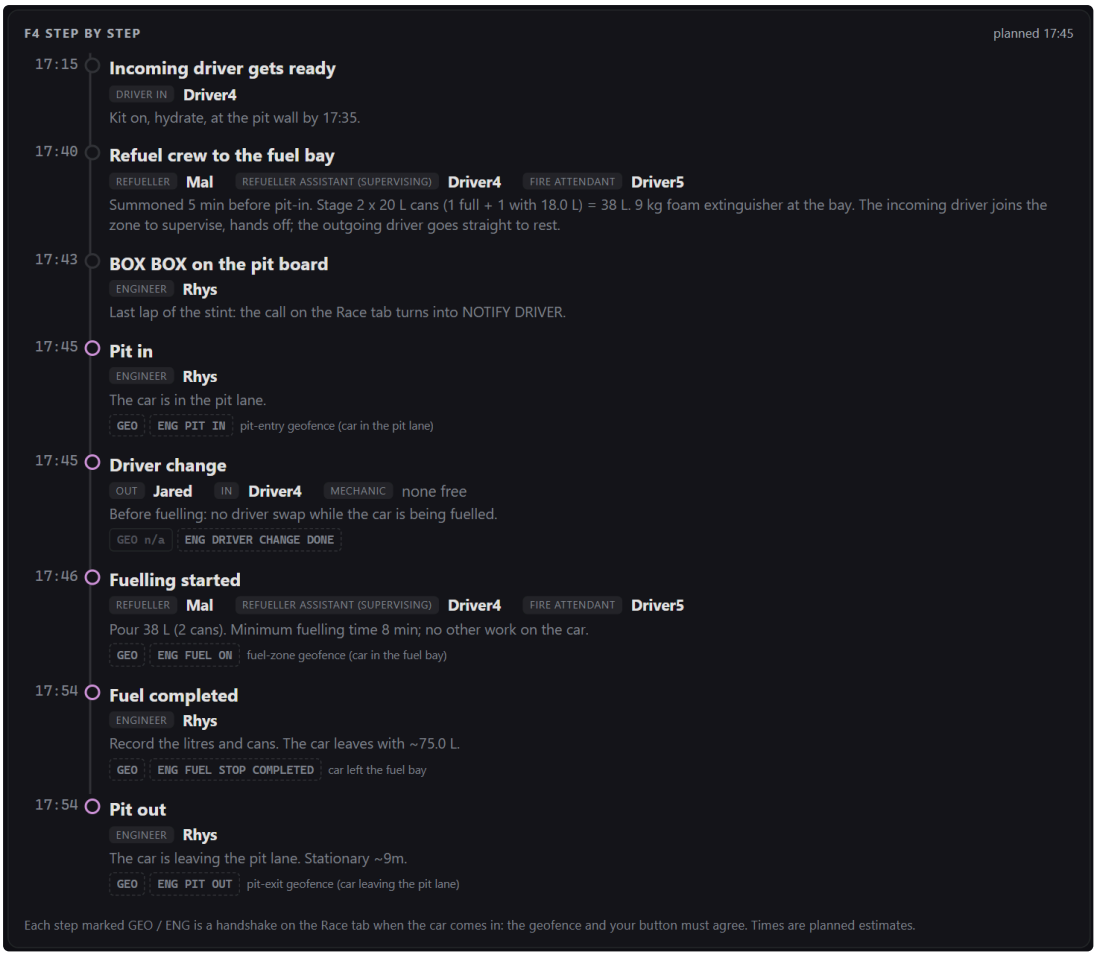


Detail of the selected fuel stop, with the litres editor

Stop detail (right card):

- litres on arrival, litres to add (*inject*), and litres on leaving;
- jerry cans to bring, for example *4 x 20 L = 3 full + 1 with 4.5 L*;
- the fuel window and the planned stationary time;
- the refuel crew by role (the incoming driver marked supervising), and the driver change;
- who else attends: pit wall engineer, radio (Raceceiver relay), lap timing, mechanic.

Crew for later stops is forecast from the shift plan at that time. The next stop uses whoever is available now.



Step by step: the procedure for the selected stop, with times, names and each step's GEO / ENG handshake

Step by step (left card): the procedure for that stop from the incoming driver getting ready and the refuel crew going to the bay (summoned 5 minutes before pit-in, cans and the 9 kg foam extinguisher staged) through BOX BOX, pit in, driver change, fuelling started, fuel completed and pit out, with planned times, the people for each step and the GEO / ENG signals that will make up its handshake on the Race tab.

Change the litres of a stop. Under the detail, set the litres with **-5**, **+5**, **Brim** or by typing. A preview shows where every later fuel stop moves, the fuel stop count and the finish fuel, before anything is changed:

- less fuel brings the next fuel stop closer; more pushes it later; every later stop re-plans;
- **Apply to the running plan** re-plans the running scenario with that amount (capped by the tank and the per-stop limit); the stop's label gets a dashed outline;
- **Back to planner litres** removes the setting for that stop; **Reset** on the Scenarios tab undoes all of them.



Preview of 10 L less at F5: the later fuel stops move earlier before you apply anything

5.8 Scenarios tab

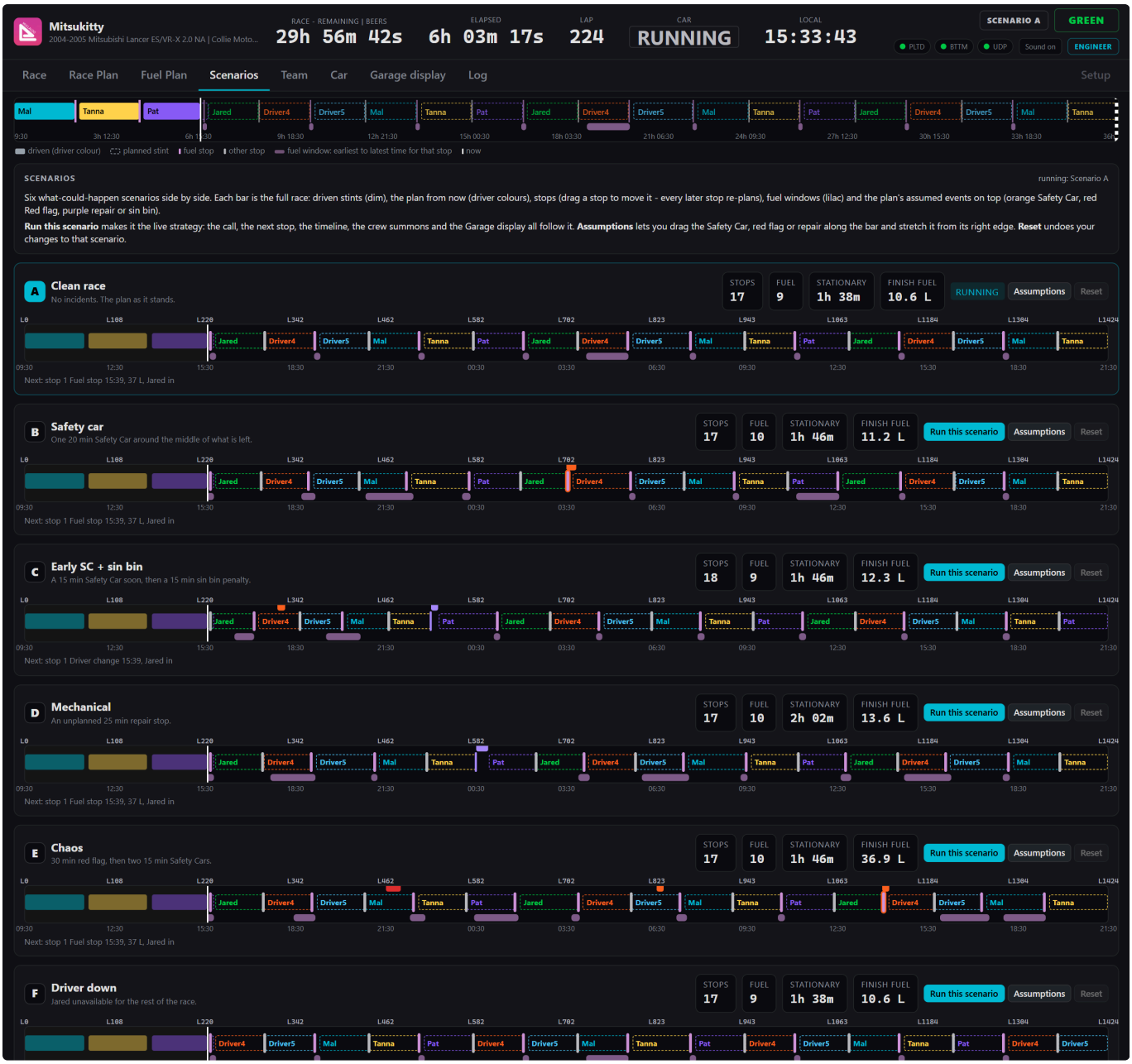


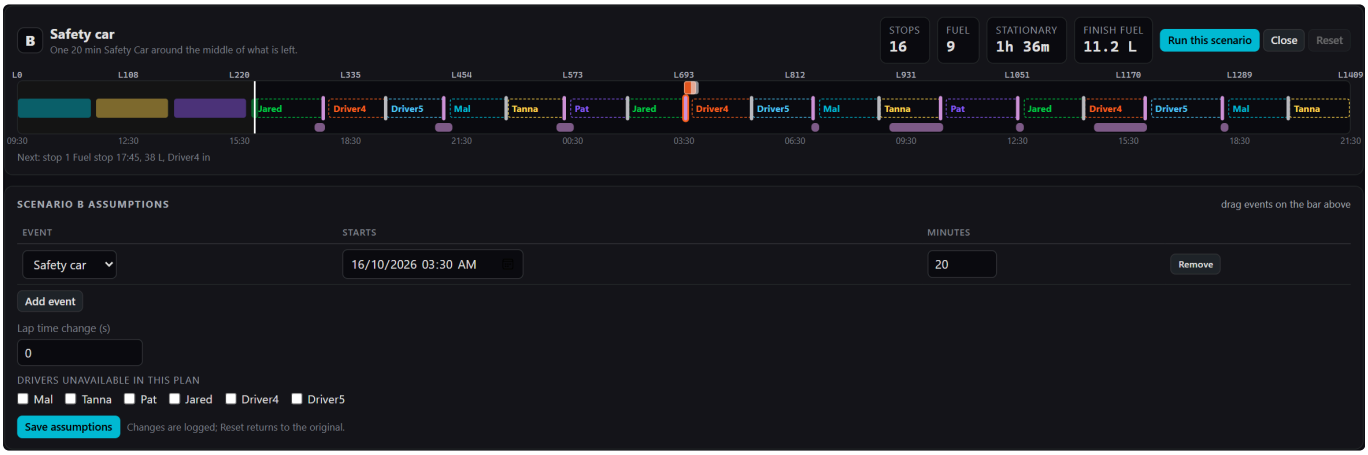
Figure 7 - Scenarios A-F, one bar per scenario

Six "what if this happens" scenarios side by side. Each starts from a template created at the moment the scenarios are created; event times are then fixed until you move them.

Scenario	Assumption
A - Clean race	No incidents: the plan as it stands.
B - Safety car	One 20 min Safety Car around the middle of what is left.
C - Early SC + sin bin	A 15 min Safety Car soon, then a 15 min sin bin penalty.
D - Mechanical	An unplanned 25 min repair stop.
E - Chaos	A 30 min red flag, then two 15 min Safety Cars.
F - Driver down	The up-next driver unavailable for the rest of the race.

Each bar is the whole race. Above it, the estimated lap number; below it, the clock time at each race hour. On the bar: driven stints dim, the plan from now in driver colours, stops as ticks (lilac = fuel), fuel windows underneath, and the scenario's events along the top (orange Safety Car, red Red flag, purple repair or sin bin). Chips give stops, fuel stops, total stationary time and finish fuel; warnings (stint over the limit, fuel below reserve) appear under the bar.

- **Box under the Safety Car.** If a Safety Car starts and a stop is due within the next 45 minutes, the scenario takes that stop under the Safety Car (orange ring) and re-plans every other stop around it. Move the Safety Car and the stops move with it.
- **Drag a stop** along the bar to move it. That stop and every stop before it are pinned (cyan ring); every later stop re-plans. Moving a stop moves the stop itself (and its driver change) to a new time.
- **Live preview while dragging:** grab a fuel stop or its (thick, lilac) fuel window and move it. The bar re-plans as you drag: the stints you shorten or lengthen, the stops after it, the stop count and the finish fuel update live, and a (!) appears if the arrangement breaks a limit. Nothing is saved until you let go.
- **Move the fuel to another stop:** drag a fuel stop and drop it onto another stop (the hint reads "drop: move the fuel to stop 2"). Both stops keep their times and driver changes; only the fuel moves.
- **Click a stop** to make it a fuel / no-fuel stop, or unpin it and the stops after it.
- **Assumptions** opens assumptions mode. The events on the bar grow handles: drag a Safety Car, red flag, repair or sin bin to another time, or drag its right edge to make it longer or shorter; everything after it cascades. The editor below the bar lists the same events with exact start times and lengths, plus a lap time change and unavailable drivers.
- **Reset** brings the scenario back to its original events and removes every dragged stop and every litres setting from the Fuel Plan. An *edited* badge shows which scenarios differ from their original.
- **Run this scenario** makes it the live strategy. The call, Next stop, Fuel Plan, timeline, refuel crew summons, prepare alerts and Garage display all follow it, and the status bar shows SCENARIO B (for example). Its events are drawn on the main timeline.



Scenario B in assumptions mode: the Safety Car (orange) can be dragged and stretched; the assumptions editor below the bars

> **Race Plan what-if vs Scenarios**
> The what-if on the Race Plan tab is a quick look on one screen and is never saved. Scenarios are saved, shared with every device, logged, and one of them is always the one running.

5.9 Team tab

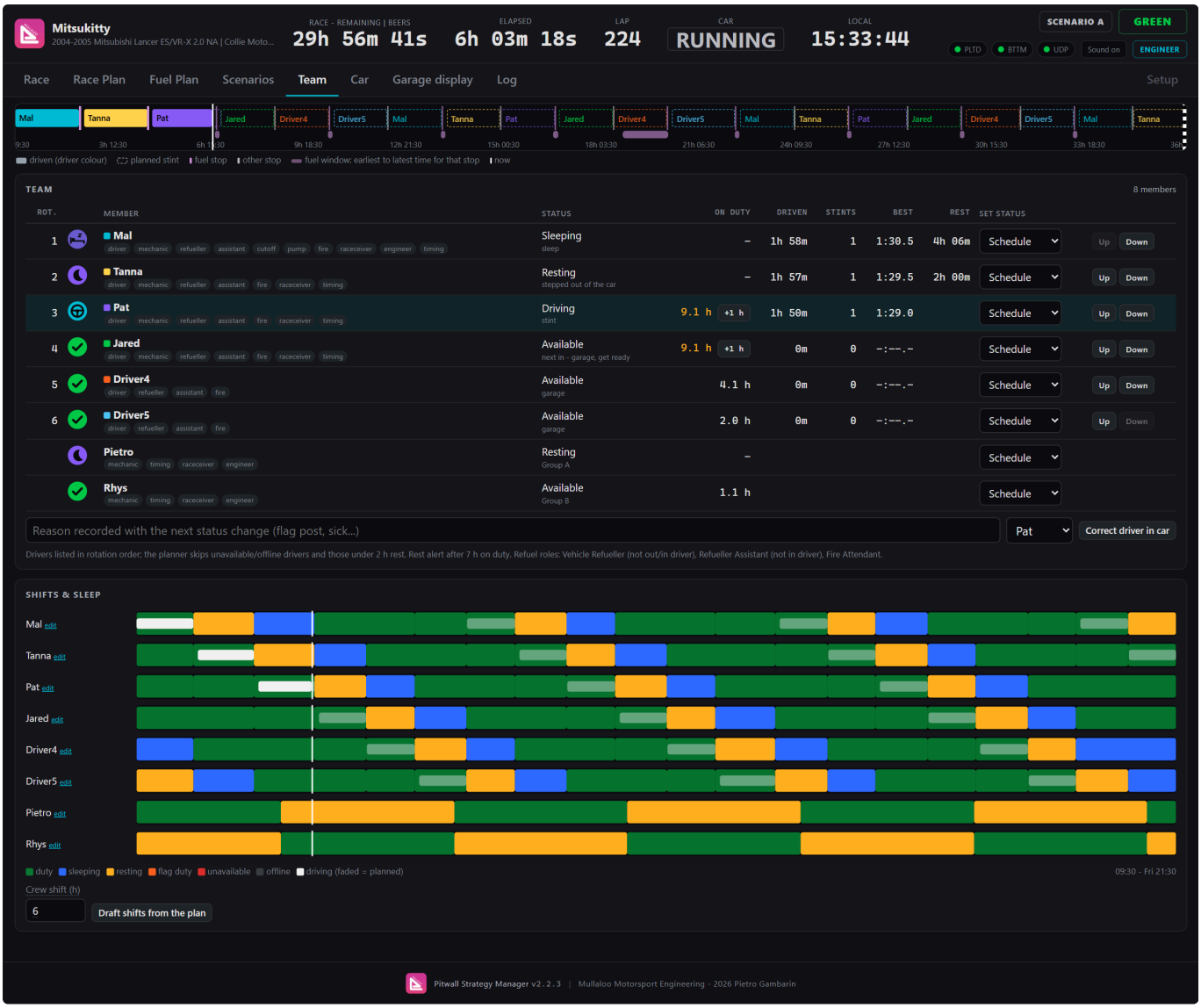
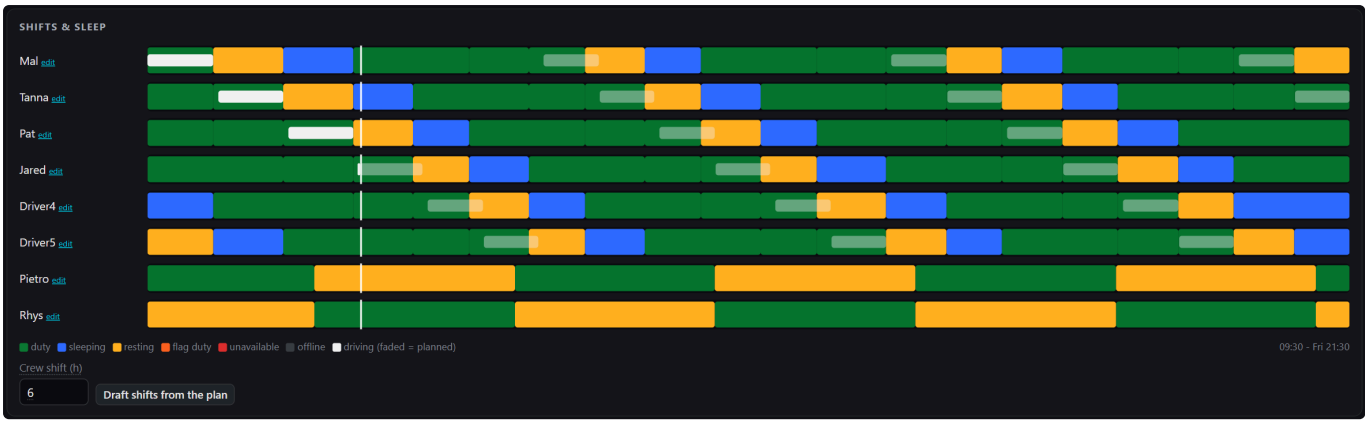


Figure 8 - Team tab: roster, then Shifts & sleep

One table for every team member. Drivers come first in rotation order (**Up**/**Down** to change it); the planner follows this order, skipping drivers who are unavailable, offline or have not rested enough. For each member: presence icon, roles, status (with what the shift plan has them doing, for example "next in - garage, get ready" or "stepped out of the car"), hours on duty (amber when rest is due, **+1 h** to override for a contingency), and for drivers time driven, stints, best lap and time since their last stint.

Status	Meaning
Available	On duty. Can be assigned to the refuel crew.
Driving	In the car (automatic).
Resting	On a break. Not assigned. The outgoing driver is set Resting automatically at pit in.
Sleeping	Asleep. Not assigned; FIX can wake them if a refuel role is empty.
Offline	Away from the track.
Flag duty	Marshalling at a flag point (enter the post in the reason box).
Unavailable	Sick, injured, or otherwise out; the planner removes a driver from the rotation.
Schedule	Follow the shift plan below (removes a manual status).



Shifts & sleep, with the high-contrast colours of 2.2.2

The **Shifts & sleep** chart shows each person's blocks over the race. The bars are tall and the colours contrast clearly NEW; the legend uses the same colours:

Colour	Block
Green	Duty
Amber	Resting
Blue	Sleeping
Orange	Flag duty
Red	Unavailable
Dark grey	Offline
White	Driving (faded = planned)

Draft shifts from the plan NEW replaces every block with a complete rota built from the plan being run (it asks first). Drivers are rostered stint by stint. During every stint:

- the driver in the car is driving, and the **next driver in** is on duty in the garage getting ready;
- the driver who has just **stepped out rests**, and the one before **sleeps** (races of 12 hours or more; shorter races rest instead);
- **at least two drivers wait in the garage** on duty, for mechanical work and the fuel crew; any extra drivers sleep.

So each driver rotates through drive, rest, sleep, garage and next in. Other crew run in two alternating groups (A/B) of the **Crew shift (h)** length for races of 12 hours or more. For every planned fuel stop the refuel crew is on duty from 5 minutes before pit-in to pit-out, chosen from who is in the garage at the summon, so the planned schedule no longer wakes anyone for a stop; when a role can only be filled by waking someone, the draft picks whoever has been called in least. Use **edit** next to a name to change individual blocks.

5.10 Car tab NEW

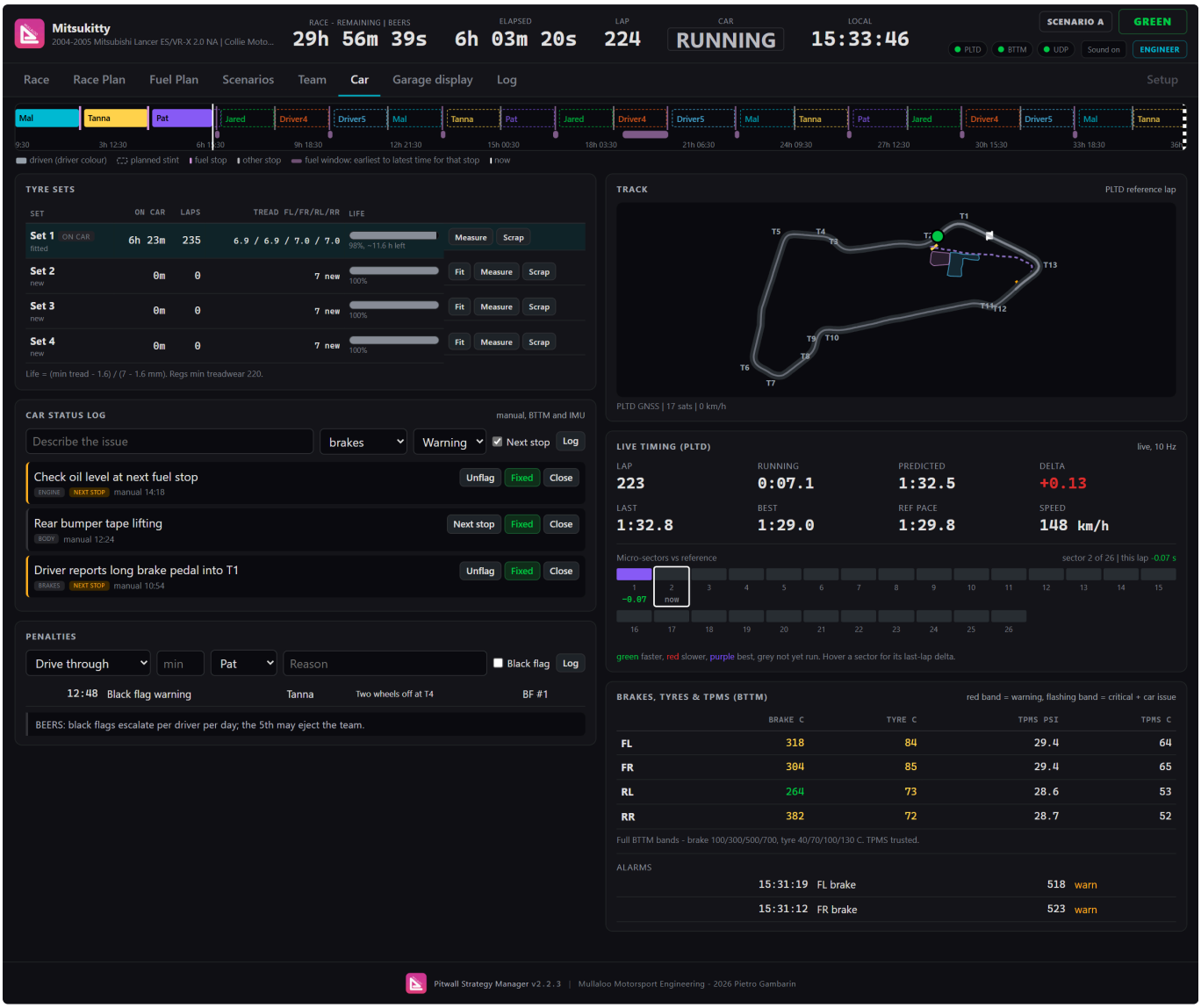


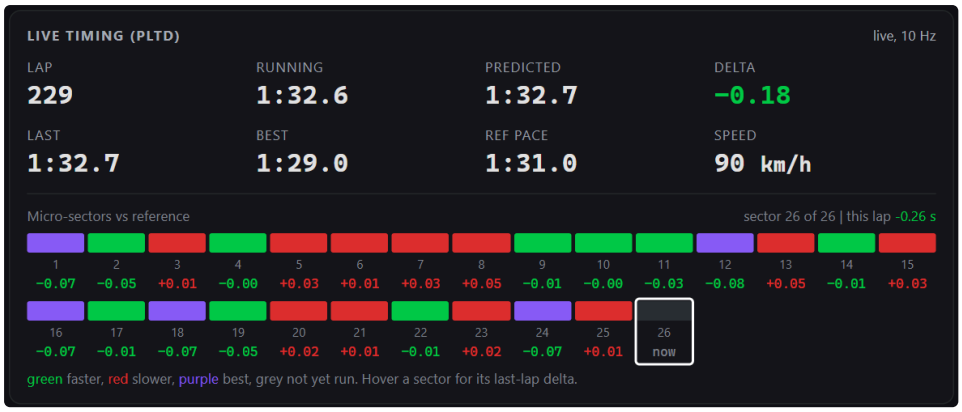
Figure 9 - Car tab: Tyre sets, Car status log and Penalties on the left; Track, Live timing and BTTM on the right

Left column, top to bottom:

- **Tyre sets** - every set for the weekend: time on the car, laps, last tread measurement and life bar. **Fit**, **Measure** (four tread depths) and **Scrap**.
- **Car status log** - every issue, manual or automatic (BTTM, IMU): category, severity, whether to deal with it at the next stop, and its history. Buttons: **Next stop**/**Unflag**, **Fixed**, **Close**.
- **Penalties** - drive-through, stop-and-go, sin bin, black-flag warnings and time penalties per driver; black flags are counted per driver per day (BEERS escalation).

Right column, top to bottom:

- **Track** - the same live GNSS map as on the Race tab.
- **Live timing (PLTD)** - lap number, running lap time, predicted lap, delta, last, best, reference pace and speed, updated live at 10 Hz.
- **Brakes, tyres & TPMS** - full BTTM detail with the device's own colour bands, thresholds and alarm history.



Live timing (PLTD) with the micro-sector grid

The Live timing card's **micro-sector grid** shows every sector of the PLTD's reference lap with its colour (green faster, red slower, purple best, grey not yet run), its number and its delta for the current lap; the sector the car is in is outlined and marked *now*. Hover a sector for its last-lap delta. Above the grid: the sector the car is in and the delta for this lap so far; the this-lap and last-lap totals follow the grid.

5.11 Garage display

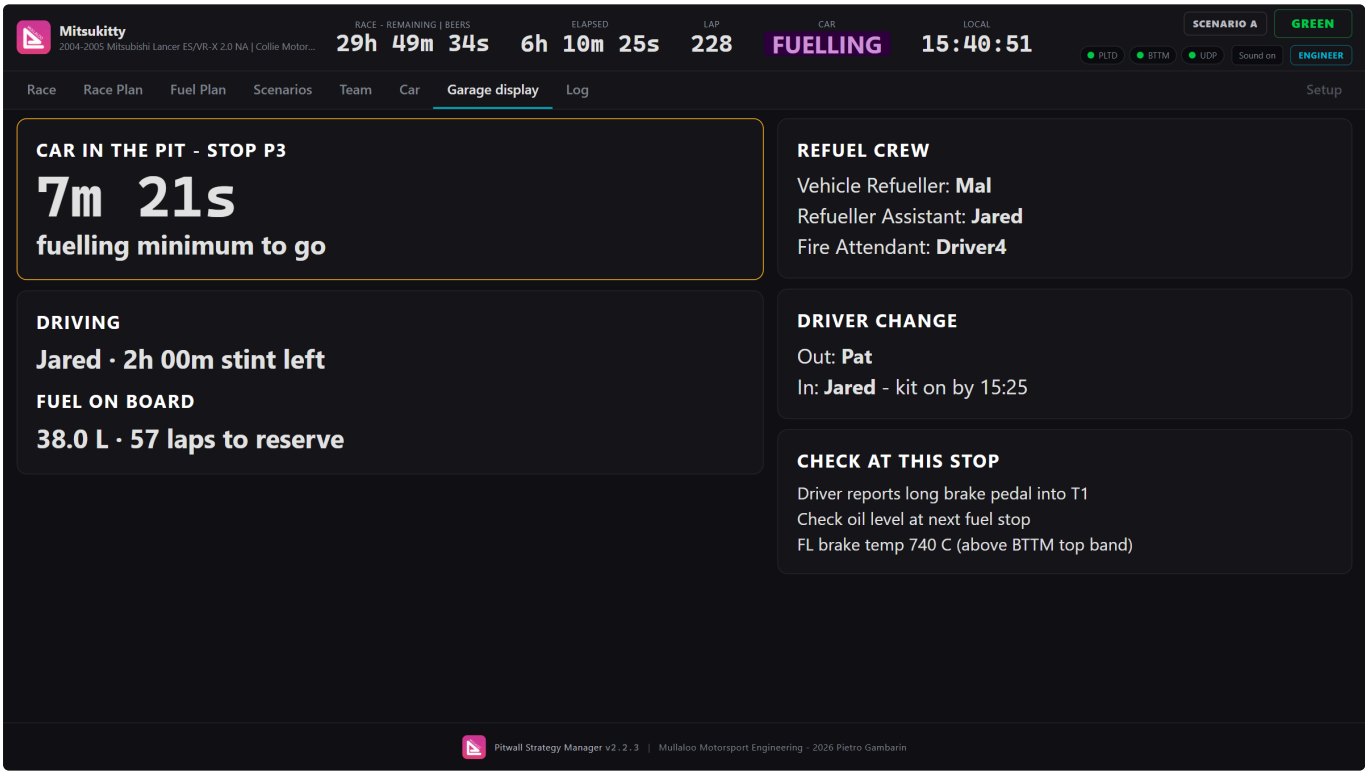


Figure 10 - Garage display during a stop

A big-type screen for the crew, meant for a TV or tablet in the garage, readable from a distance, with large white titles. An unassigned refuel role shows a **FIX** button on the engineer's screen. Between stops it shows the countdown to the next stop, its type and fuel, the driver and fuel on board. When the car is in, it shows the fuelling minimum counting down. Alongside: the refuel crew by name, the driver change (with the time to have kit on), jobs to check at this stop, and a warning when refuelling is closed by the flag.

5.12 Log tab

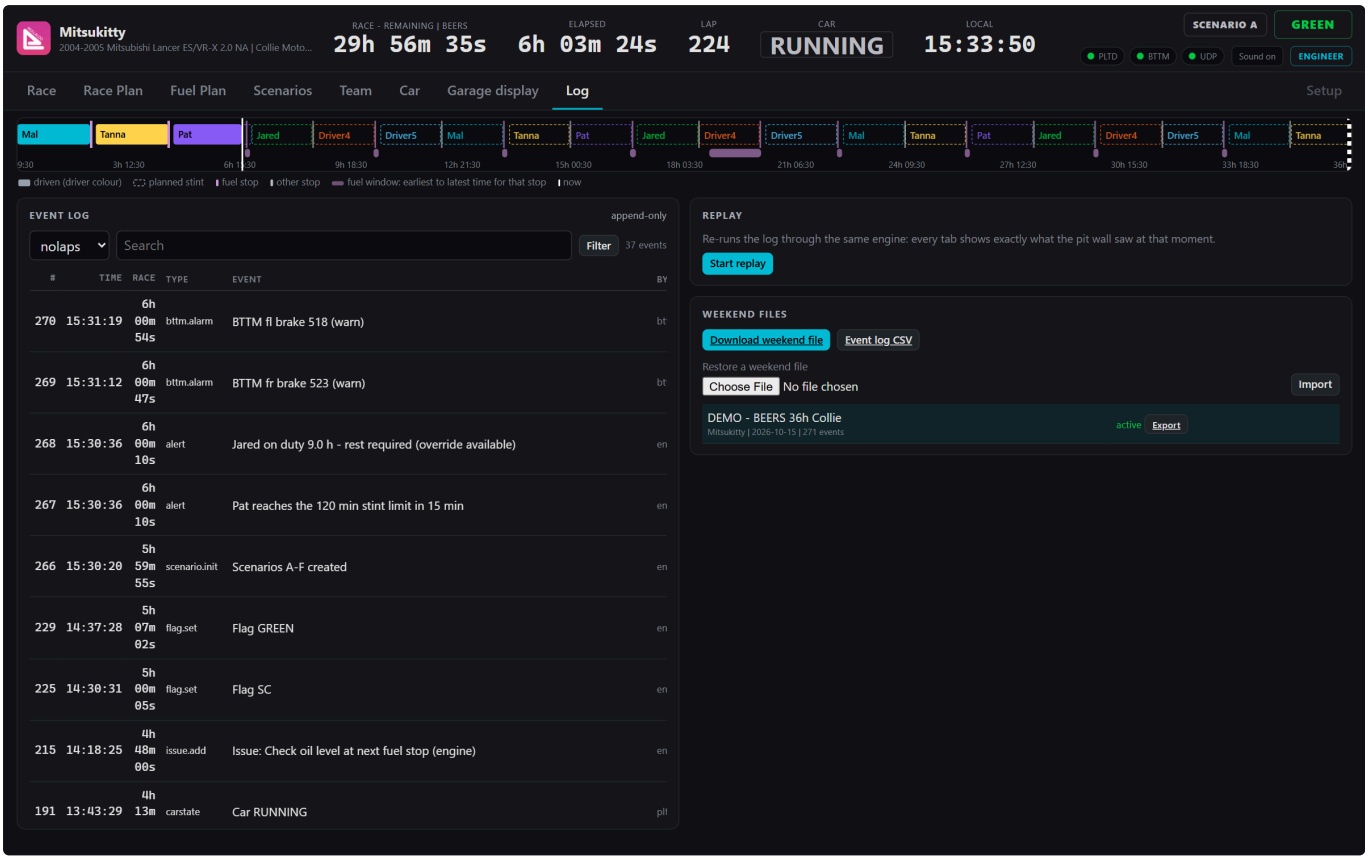


Figure 11 - Log tab

- **Event log** - every change ever made, in order, with clock and race time and who made it. Filter by type or search text. New entries include the stop handshake steps and "Stint extended +1 lap - stop N moved to ~hh:mm".
- **Replay** - re-runs the log through the application: every tab shows what the pit wall saw at that moment. Slider, Play at 1x to 1200x, jump 10 minutes, **Back to live**.
- **Weekend files** - download the weekend file (setup and complete log in one file), the event log as CSV, restore (import) a weekend file, and open other weekends.

5.13 Setup

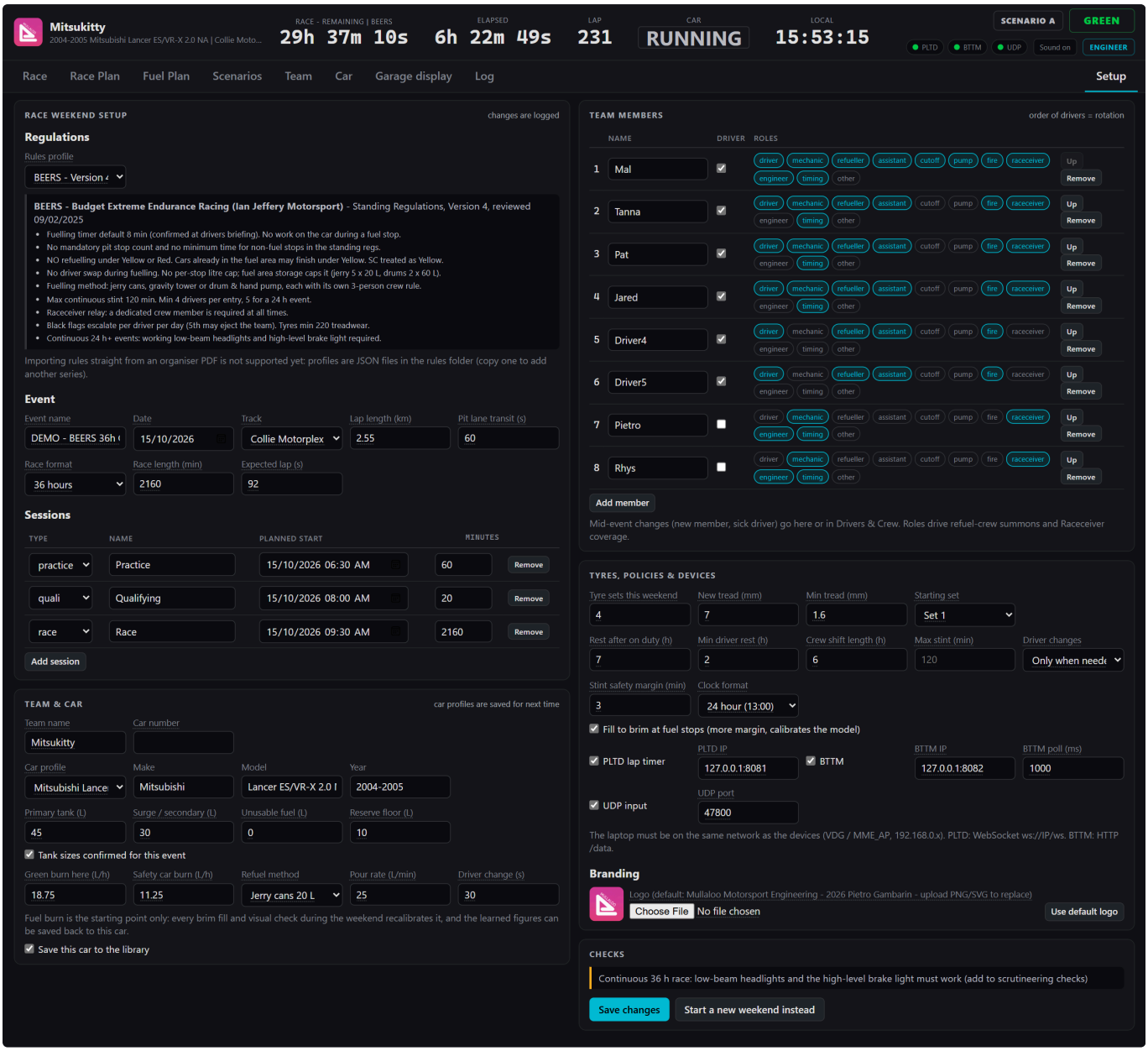
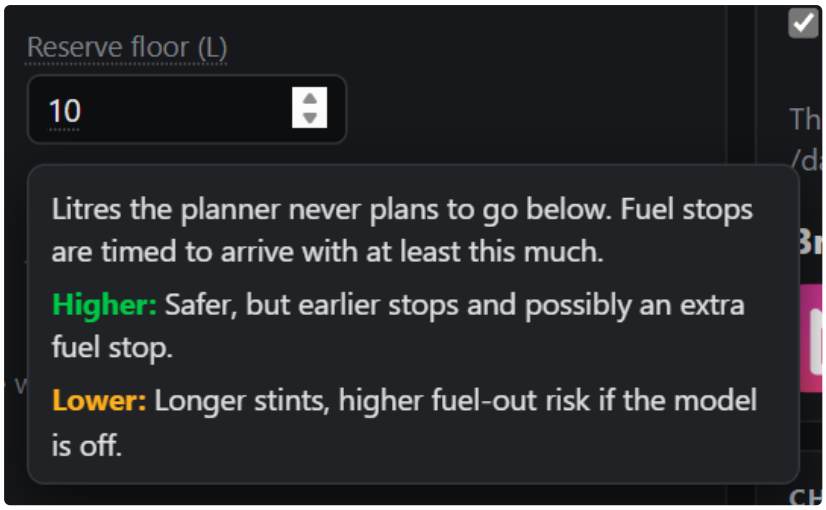


Figure 12 - Setup

Opened from **Setup** at the right of the tab bar. On the first start it is the only screen, and it creates the race weekend. Later it edits the weekend; every change is logged.



Every field has a tooltip: what it is, and what raising or lowering it does

Section	Fields
Regulations	Rules profile (B4B Events or BEERS) with its version and key rules.
Event	Name, date, track (the Collie Motorplex preset includes the long-circuit outline and turn numbers), lap length (only for distance races), pit lane transit time, race format (7 h, 12 h, 24 h, 36 h, 1000 km, custom), race length, expected lap time.
Sessions	Practice, qualifying, race (and more) with planned start and length.
Team & car	Team name, number; car profile from the library; tank sizes and unusable fuel; reserve floor; **Tank sizes confirmed**; baseline green and Safety Car burn for this track; refuel method; pour rate; driver change time; save car to library.
Team members	Name, driver yes/no, roles (driver, mechanic, refueller, assistant, cutoff, pump, fire, raceceiver, engineer, timing, other). Drivers' order is the rotation.
Tyres, policies & devices	Tyre sets, new and minimum tread; rest after hours on duty; minimum driver rest; crew shift length; max stint (at or below the regulations); when to change drivers; stint safety margin; fill to brim; clock format (24 h or 12 h am/pm); PLTD, BTTM and UDP addresses.
Branding	Logo: the Mullaloo Motorsport Engineering logo is built in; upload a PNG/SVG only to replace it, **Use default logo** to go back.
Checks	Warnings before you race: too few drivers for the format, tank over the regulation maximum, tank sizes not confirmed, refuel roles nobody holds, no Raceceiver relay (BEERS), headlights for 24 h+ races.

5.14 Track layout - Collie Motorplex long circuit

The Collie Motorplex preset is the long circuit: 2.55 km, 13 turns, run anti-clockwise, start/finish on the straight between T13 and T1. Its geometry is real:

- the circuit centreline was fitted to satellite imagery of the venue (true shape and orientation, north up) using the official layout below for the turn order, and scaled to 2.55 km;
- the pit lane, the pits area, the fuel area, the pit entry and exit lines and the start/finish line were taken from the team's own mark-ups of the pit complex. The pit lane is about 460 m from entry to exit and runs alongside the start/finish straight; the pits area comes before the fuel area, then the exit.

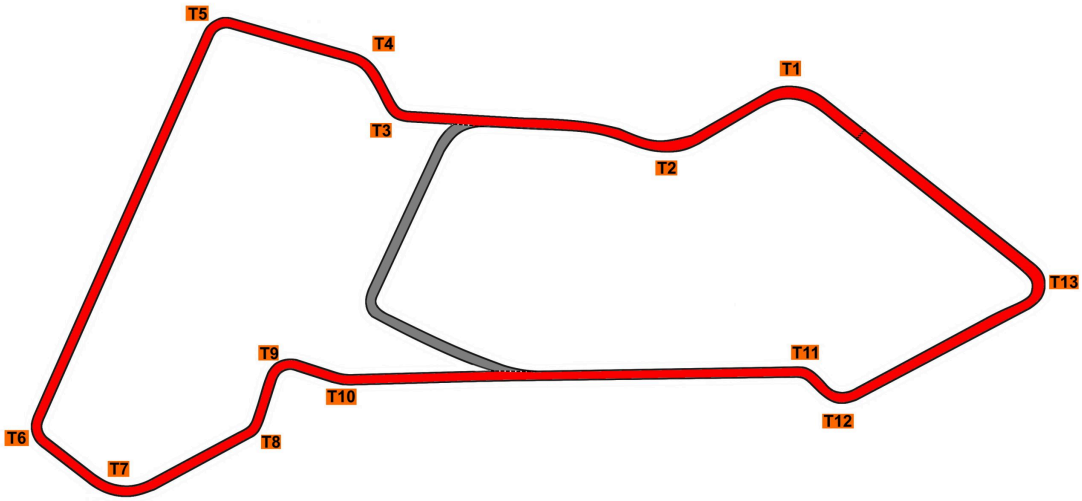


Figure 13 - Collie Motorplex long circuit, official layout (red). The grey link is the short-circuit connection and is not part of the long circuit.

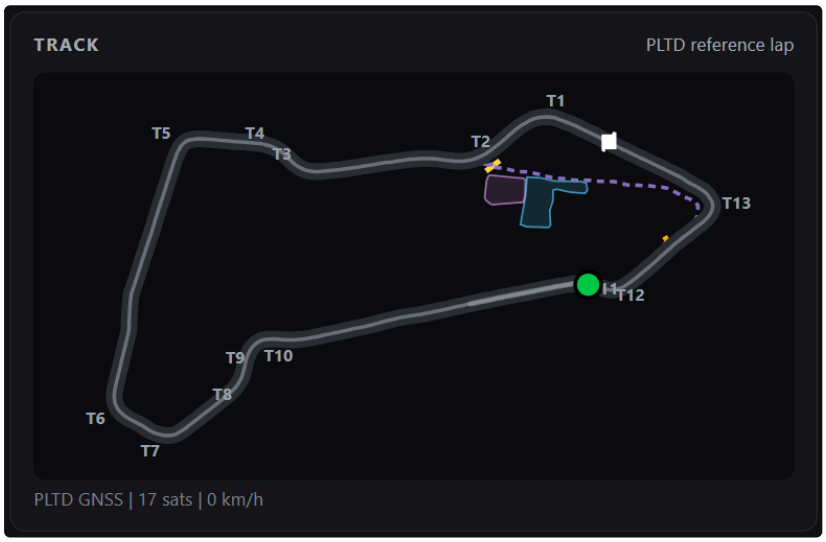


Figure 14 - The circuit on the map: turn numbers, start/finish line, pit lane (dashed), pits area (blue), fuel area (lilac), pit entry and exit lines

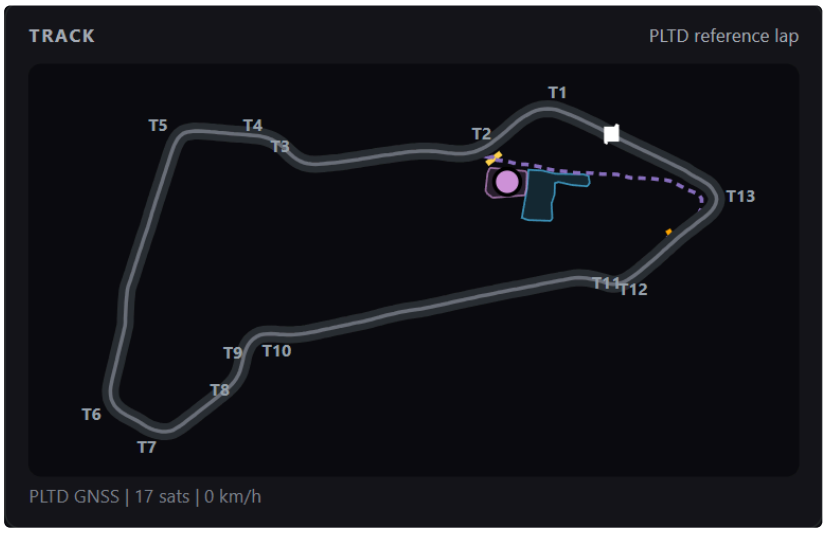


Figure 15 - The car in the fuel area during a stop

> The map uses the preset until the PLTD has recorded its own reference lap, then the PLTD's lap (exact for the car's GPS). Turn numbers and the pit lane stay on the map whenever they line up with the drawn circuit. The preset's absolute GPS position is approximate (shape, scale and orientation are true). At the track the PLTD's recorded lap and its stored geofences are what the app uses.

6. Running a real race weekend

6.1 Before the weekend

1. Start the server (with no weekend active, Setup opens; otherwise **Setup - Start a new weekend instead**).
2. Choose the regulations, track, format and sessions. Enter team, car and tank sizes; tick **Tank sizes confirmed** once checked on the car.
3. Enter every team member and tag their roles. For jerry-can fuelling you need people tagged `refueller`, `assistant` and `fire`. Remember that the outgoing driver is never on the crew and the incoming driver takes the assistant role (supervising), so tag at least three or four drivers or crew with each fuel role; for BEERS also `raceceiver`.
4. Clear all warnings in **Checks**, then **Create race weekend**. Upload the logo once if you do not want the built-in one.
5. In **Team**, press **Draft shifts from the plan** and adjust blocks as needed. Open **Fuel Plan** and check the crew forecast for each fuel stop.

6.2 At the track: connect the car

Join the laptop to the car network (VDG / MME_AP, 192.168.0.x). Read the IP addresses on the PLTD and BTTM screens and enter them in **Setup - Tyres, policies & devices**, then **Save changes**. Starting from the demo weekend, these are the fields to change (top to bottom):

Description	Current setting	Target setting
PLTD lap timer	ticked	no change
PLTD IP	127.0.0.1:8081 (simulator)	IP shown on the PLTD screen, e.g. 192.168.0.41
BTTM	ticked	no change
BTTM IP	127.0.0.1:8082 (simulator)	IP shown on the BTTM screen, e.g. 192.168.0.40
BTTM poll (ms)	1000	no change
UDP input	ticked	no change
UDP port	47800	no change

The PLTD and BTTM indicators in the status bar turn green within seconds. The track map appears once the PLTD has recorded its reference lap. Close the simulator if it is still running.

6.3 Starting a session

In **Race control** select the session, enter the actual fuel in the car, the starting driver and the tyre set, and press **Green flag - start session** when the session goes green. Practice and qualifying are recorded too: their laps and fuel measurements help the model before the race.

6.4 During the race - routine

- Glance at the call and the Attention list; acknowledge alerts once handled.
- Set the flag in Race control whenever race control changes it; the call accounts for refuelling rules under each flag.
- About 30 and 15 minutes before a driver change, the next driver gets a "prepare" alert; 5 minutes before a fuel stop the refuel crew is summoned by name.
- At every stop: confirm the handshake steps and record the fuel (litres, brim, driver in, fixes). This is what keeps fuel and driver times accurate.

- Log a gauge reading whenever someone reads the fuel gauge.
- Keep crew statuses current (Resting, Sleeping, Flag duty, Unavailable); assignments update immediately.
- Use the **Garage display** on a TV and the **Fuel Plan** on the crew chief's tablet.

6.5 Pit stop workflow

Moment	What happens	Engineer action
Stop near	From 5 min before the planned stop the call counts laps: BOX IN 3 LAPS, 2, 1. On the last lap: NOTIFY DRIVER: BOX BOX (plus a notification). Refuel crew summoned 5 min before pit-in; the incoming driver joins them in the fuel zone.	Press **Send BOX BOX to the driver** in the call card (or BOX BOX in Race control). The call then shows when it was sent and acknowledged. To keep the car out, press **+1 lap** instead.
Pit entry	PIT ENTRY geofence opens the stop (GEO); the handshake card appears; pit lane timer starts. The outgoing driver is set Resting.	**PIT IN** (or **Not a stop**). Brief the crew from "Check at this stop".
Driver out	Outgoing driver leaves the car before fuelling (nobody in the car while fuelling) and goes straight to rest.	Check the driver row, **DRIVER CHANGE DONE**.
Fuelling	Fuel bay geofence (GEO): fuelling timer against the minimum in the Pit lane card. The incoming driver supervises, hands off; the refueller and fire attendant pour and stand by.	**FUEL ON** when pouring starts. Fill in the Fuel stop card. No other work on the car.
Fuel done	Car leaves the fuel zone (GEO). Incoming driver in after fuelling.	**FUEL STOP COMPLETED**.
Pit exit	PIT EXIT geofence: stop closed and checked (fuel time, mandatory minimum); plan recalculated. Anything not confirmed is applied from the plan (AUTO).	**PIT OUT**, then check the new next stop.

> **If the PLTD misses a geofence**
> Use **Override car state** under the Pit lane card (or the manual car state buttons in Next stop when the PLTD is offline). The stop timers follow these buttons exactly as they follow the PLTD.

6.6 After the race

Press **Chequered**. In **Race Plan - Fuel model**, press **Save learned burn to this car** to keep the measured consumption for the next event at this track. In **Log** download the weekend file and the CSV for the debrief, and back up the `data` folder.

7. How the strategy works

7.1 Fuel model

Burn starts from the car's baseline for the track (green and Safety Car litres per hour). Every lap is classified by its time against the live reference pace: push, green, slow, Safety Car or pit lap; stationary time in the pits burns nothing. Each brim fill or gauge reading is a measurement of the fuel actually used since the previous one; a regression then refits the green and Safety Car rates, with the baseline acting as a starting point. A stint that used more than 10% more or less than predicted raises an alert. The estimate shows its uncertainty, which grows with time since the last measurement.

7.2 Planner

From the current moment to the flag, the planner searches the number of stops, how many of them are fuel stops and where they fall, and picks the plan with the least total time stationary while respecting:

- fuel range and the reserve floor;
- the maximum continuous stint (minus the safety margin);
- the maximum litres per stop (regulation cap or fuel-area storage);
- mandatory stops and their minimum duration (B4B);
- the minimum fuel stop time and pour rate;
- driver rotation, availability and minimum rest.

On long races with short fuel range (for example the BEERS 36 h), this typically alternates driver-only stops with fuel stops rather than fuelling at every stop, which saves a large amount of stationary time.

7.3 Scenarios and fixed stops

A scenario is the planner plus assumptions. Safety Car windows lower the burn rate (and a stop due within 45 minutes is taken under the Safety Car), red flags stop the burn, repair stops and sin bins are forced stops of fixed length, and unavailable drivers are removed from the rotation. Stops you drag are pinned: the planner keeps them where you put them, with their fuel and driver choice, and searches the best plan only for the time after the last pinned stop (and in front of any forced event stop). Litres set on the Fuel Plan are kept the same way: that stop gets that amount and everything after it re-plans. Warnings tell you when a pinned arrangement breaks a limit.

7.4 Windows and calls

A fuel window opens at the earliest time the car could stop, fill up and still reach the next fuel stop (or the flag), and closes at the latest time before the reserve floor or the stint limit. The call is recalculated every second from the plan, the fuel estimate, the stint time and the flag; Appendix B lists every call.

7.5 Driver time and +1 lap

A driver's continuous time stops when the car stops in the pits. Once the driver change is confirmed, the Now column shows the new driver as "In the car - timer starts at pit out", with the outgoing driver's final time beside it; the new driver's clock starts when the car leaves the pit lane. While the car is in the pits no stint-limit or fuel-window alerts are raised. A driver who stays in the car through a stop keeps counting continuous time from where it paused.

+1 lap NEW holds the called stop one reference lap later (never closer than the next minute) in the running plan and every scenario. It is allowed only if the fuel on arrival stays at or above the reserve floor and the stint stays within the maximum, counting to the middle of the pit lane. The stop's fuel grows by one lap's burn; the planned driver change is kept. The extension ends when the car next enters the pits.

7.6 Crew, refuel crew and shifts NEW

- **Presence states:** Available, Driving, Resting, Sleeping, Offline, Flag duty, Unavailable, each with a reason. A manual status overrides the shift plan until set back to **Schedule**.
- **Rest:** rest alert after 7 h on duty (Setup), with a 1 h contingency override. Drivers need the minimum driver rest (2 h by default) before the planner puts them back in the car.
- **Prepare alerts** for the next driver at 30 and 15 minutes.
- **Refuel crew** summoned by name 5 minutes before a fuel stop. Only members **tagged** with the role and currently **Available** are assigned, the most constrained role first, preferring non-drivers.
- **The golden rule at a fuel stop:** the outgoing driver steps out and goes to rest - never on the refuel crew (set Resting automatically from pit in). The incoming driver enters the fuel zone as **Refueller Assistant, supervising only** (hands off). Pouring and the extinguisher go to the other available members tagged for them.
- If nobody qualifies, the slot shows **UNASSIGNED** with the reason; available untagged crew are listed as suggestions and tagged crew who are resting, sleeping, offline or on flag duty can be woken with **FIX**, never assigned silently.
- **Shift draft:** see section 5.9. Drivers rotate drive - rest - sleep - garage - next in, at least two drivers are in the garage during every stint, and the fuel crew for each planned fuel stop comes from them.

7.7 Regulation profiles

Rule	B4B Events (Mar 2026 + 7 h supplementary)	BEERS (Standing Regulations V4, 09/02/2025)
Fuel stop minimum	5 min	8 min (confirmed at the briefing)
Mandatory stops	5 stops of at least 5 min	None
Fuel per stop	Max 80 L	Limited by fuel-area storage (jerry cans 5 x 20 L, drums 2 x 60 L)
Refuelling under Safety Car	Allowed	Not allowed (treated as Yellow)
Refuelling under Red	Not allowed	Not allowed
Max continuous stint	120 min	120 min
Drivers	3 to 6	Min 4, min 5 for 24 h+
Other	Contact = 15 min sin bin	Raceceiver relay required at all times; black flags escalate per driver per day; headlights for 24 h+; no driver swap during fuelling

Profiles are JSON files in the `rules` folder. Importing directly from an organiser's PDF is not supported in this release.

8. Devices and network

Device	Connection	Used for
PLTD lap timer (v0.17)	WebSocket to `ws://PLTD-IP/ws` (10 Hz live data, flag state, pit board shown / acknowledged); HTTP `/data` (completed lap time, position at 2 Hz), `/refpath`, `/geozones`, `/sf/get`	Laps, live timing and micro-sectors, car state from geofences (stop timers and the GEO half of the handshake), position and track map, pit board messages and driver acknowledgements, flags shown on the dash
BTTM (v4.5)	HTTP `/data` polled every second, `/config` for thresholds	Brake and tyre temperatures, TPMS, alarms; GPS position as a fallback for the map
Other devices	JSON over UDP, port 47800	Simulator, future devices (e.g. an IMU impact sensor)

UDP messages are one JSON object per datagram:

```
{"t":"lap","n":12,"ms":91234,"pitted":false}      {"t":"carstate","cs":"PITS"}
{"t":"flag","flag":"SC"}                          {"t":"fuel","litres":32.5}
{"t":"impact","g":4.2,"lat":-33.36,"lon":116.15}
{"t":"issue","cat":"engine","sev":"warn","text":"Oil temp high"}
```

BTTM alarms: a value in the BTTM red band is logged as a warning; above the top (flashing) band it is critical and automatically creates a car issue flagged for the next stop. Each alarm level repeats at most every 5 minutes per corner.

9. Data, backup and recovery

Path (inside the data folder)	Content
`settings.json`	Port, engineer hosts, last weekend (resumed automatically)
`cars\`	Car library with learned fuel profiles per track
`events\<id>\config.json`	The weekend as created
`events\<id>\events.jsonl`	Every change, one line each: the race log and replay source
`events\<id>\car_config.json`, `driver_schedule.json`	Readable snapshots
`branding\`	Your logo

- Changes are written to disk the moment they are made; a power cut loses at most the last change.
- The weekend file (**Log - Weekend files - Download**) contains the setup and the complete log; **Import** restores it on any machine.
- Copy the `data` folder to a USB stick or a second laptop at least once per session during long events.

10. Troubleshooting

Symptom	Cause and fix
Console window closes immediately	Node.js is not installed or too old. Install Node.js 18+ (LTS).
"Reconnecting to pit wall server" banner	The server stopped or the network dropped. Restart `start-pitwall.bat`; the weekend resumes. On other devices check the Wi-Fi.
Other devices cannot open the page	Firewall blocked Node.js (allow on Private networks), wrong address (use the one printed in the console), or devices on a different network.
VIEW ONLY on the laptop	You opened it via another machine's address. Use `http://localhost:8080` on the host.
Race tab columns stacked	The window is narrower than 1400 px. Maximise the browser or zoom out (Ctrl and minus).
PLTD or BTTM indicator red	Wrong IP, laptop not on the car network, or device off. Check the IP on the device screen and in Setup.
PLTD amber ("stale")	Connected but no live data for 3 s: weak Wi-Fi or the device is busy.
No track map / no micro-sectors	The PLTD has no reference lap yet; it records one after the first clean lap. The live trail is drawn meanwhile.
Handshake step stays amber	The geofence saw the step and you have not confirmed it. Press the step's button, or leave it: it becomes AUTO once the car moves on.
+1 lap greyed out	The extra lap would bring the fuel below the reserve or the stint over the maximum; the tooltip gives the reason.
Refuel crew UNASSIGNED	Nobody is tagged with that role, or everyone tagged is resting, sleeping, offline, driving or barred by the rules (outgoing / incoming driver). The reason and possible helpers are shown; press FIX, or tag more people in Setup.
Fuel estimate drifting	No measurement for a long time. Log a gauge reading or tick **Filled to brim** at the next fuel stop.
Demo times look wrong	The demo was built a while ago. Stop the server, run `make-demo.bat` again.
Simulator: "address in use"	Another simulator is still running. Close it, or start with `--pltd` / `--bttm` on other ports and update Setup.
Changes not accepted	Replay is active (press **Back to live**) or the device is view-only.

11. Known limitations of this test release

- Tested against the simulator. The device adapters follow the PLTD v0.17 and BTTM v4.5 firmware exactly but have not yet been used with the real devices on a live car.
- Regulation profiles cannot be imported from PDF; they are JSON files.
- Impacts can only be received over UDP; PLTD v0.17 does not send them.
- GPS positions are live only and not stored in the log, so replay has no map.
- The race clock keeps running under Red flags (profile setting).
- One server at a time: versions 1.x and 2.x use the same port and the same data folder.
- Under a red flag the plans assume the car is stopped with the clock running; driver continuous time keeps counting.
- Crew for fuel stops after the next one is a forecast from the shift plan; the real crew is assigned from who is Available at the 5-minute summon.
- The Collie preset's absolute GPS position is approximate; its shape, pit lane and zones come from satellite imagery and the team mark-ups. The PLTD's recorded lap and geofences replace them at the track.

12. Reporting feedback

When something is wrong or unclear, send the following. The weekend file and the CSV let the problem be replayed exactly.

```
Pitwall Strategy Manager v2.2.3 - test report
Tester:
Date / time of the problem (laptop clock):
Weekend: demo / real (name):
Screen and panel:
What I did:
What I expected:
What happened:
Screenshot: attached
Weekend file (.pitwall.json) and event log CSV: attached (Log tab - Weekend files)
```

Useful things to test: the full pit stop handshake, confirmed and unconfirmed; +1 lap on the last lap; changing the litres of a fuel stop on the Fuel Plan; a driver marked Unavailable mid-race; crew going to rest and the refuel crew re-assigning; drafting shifts and checking who is called for each fuel stop; a Safety Car under both rule sets; restarting the server mid-race; two or three viewing devices at once; a long (several hour) run with the simulator.

Appendix A - Glossary

Term	Meaning
Brim fill	Filling the tank until full. Gives an exact measurement of fuel used.
Call	The single instruction shown at the top of the Now column.
ENG	The engineer's half of a stop handshake: the step's button was pressed.
Fuel window	Earliest to latest time a fuel stop can be made without breaking fuel or stint limits.
GEO	The geofence half of a stop handshake: the PLTD saw the car do the step.
Geofence	A zone stored in the PLTD (pit entry line, pits, fuel bay, pit exit line); crossing it changes the car state.
Handshake	A stop step confirmed by both the geofence (GEO) and the engineer (ENG).
Mandatory stop	A stop the regulations require (B4B: at least 5 stops of at least 5 minutes).
Micro-sector	One of the short sections the PLTD splits its reference lap into; each is timed against the reference.
Up next	The next driver to get in the car.
PLTD	The MME lap timer and pit board display in the car.
BTTM	The MME brake and tyre temperature monitor, with TPMS.
Raceceiver	Race-control radio receiver; BEERS requires a crew member relaying it at all times.
Reference pace	Median of recent clean green laps.
Reserve floor	Fuel the plan never goes below.
Stint	Continuous time one driver spends driving.
Stint extension	+1 lap: the called stop moved one lap later by the engineer.
Stationary time	Time the car is stopped in the box.
What-if	A scenario calculated on screen only; never saved.

Appendix B - Calls and alerts

B.1 Calls

Call	When
PRE-RACE	Race not started; the plan is built from the planned start.
PRACTICE / QUALIFYING	A non-race session is shown.
STAY OUT	Normal running; shows minutes and laps to the next stop.
WINDOW OPEN	The next fuel stop's window is open; shows when it closes.
BOX IN N LAPS	Less than 5 minutes left to the planned stop (the end of the driver's allocated stint): laps to go.
NOTIFY DRIVER: BOX BOX	The planned stop is at the end of this lap and the driver has not been told yet. Send BOX BOX from the call card, or press +1 lap to keep the driver out one more lap.
BOX THIS LAP (amber)	Last lap, BOX BOX already sent; shows when it was sent and whether the driver acknowledged. +1 lap is still available.
BOX THIS LAP (red)	Fuel at reserve, or the driver at the stint limit.
BOX NOW (caution)	Yellow / Safety Car and a stop is near: stopping under caution loses less time (only where refuelling is allowed under that flag).
STAY OUT - refuelling closed	A fuel stop is near but the flag forbids refuelling.
IN PITS	The car is in the pit lane.
RED FLAG	Red flag: pit lane, engine off.
CHEQUERED	Race finished.

B.2 Alerts

Alert	Level	Action
Notify driver to BOX BOX	Warning	Send BOX BOX on the pit board: the stop is at the end of this lap.
Fuel at reserve	Critical	Box this lap.
Projected finish / stop below reserve	Critical	Review the plan; consider saving fuel or an extra stop.
Consumption +/-N% vs model	Warning / critical	The model has recalibrated; check the new plan.
Fuel window OPEN / closes in 5 min	Warning / critical	Prepare or make the fuel stop.
Refuel crew to the fuel bay	Warning (critical if a role is unassigned)	Crew to the fuel bay; fix any UNASSIGNED role.
Refuelling CLOSED under flag	Warning	Hold the fuel stop until the flag allows it.
Driver reaches stint limit in 30 / 15 / 5 min; over max stint	Info / warning / critical	Plan the driver change.
Prepare (next driver) at 30 / 15 min	Info / warning	Next driver: kit on, hydrate, ready.
On duty N h - rest required	Warning	Send the person to rest, or override for 1 h.

No Raceceiver relay on duty (BEERS)	Critical	Put a raceceiver-tagged member on duty.
Check at this stop	Warning	Jobs listed for the stop in progress.
Stop fuel time under the minimum	Critical	Possible penalty: inform the team manager.
Stop does NOT count toward mandatory stops (B4B)	Warning	Another stop of the minimum length is needed.